

Chapter 2

Rational Curves & Bend and Break

Revised chapter corresponding to original pp. 97–157.

Original numbering and proof order retained.

The supplied revisions of the applications and cone theorem are incorporated.

The accompanying correction report gives the original page locations.

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2.1 Parametrizing Morphisms

2.1A Rational Curves on the Exceptional Locus

The more rational curves X contains, the more complicated the birational geometry of X .

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Rational curves play an important role in classifying algebraic varieties. For example, over an algebraically closed field of characteristic zero, a smooth projective surface has a minimal surface X_m with no (-1) -curves, as in Corollary 1.4.34. There are two cases:

- either X is not covered by rational curves, in which case its minimal surface is unique and K_{X_m} is nef;
- or X is covered by rational curves, in which case minimal surfaces need not be unique, and their canonical divisors are not nef.

The characteristic-zero assumption in this introductory classification is relevant; it is not a statement about all surfaces in positive characteristic.

[Bea78, II, 14(b)] gives a simple example for the second case. Start with $\mathbb{P}^2$ blown up at two distinct points to get $\widehat{Q}$, and then blow down the strict transform of the line joining them. This gives the smooth quadric surface $\mathbb{P}^1 \times \mathbb{P}^1$. Thus $\mathbb{P}^2$ and $\mathbb{P}^1 \times \mathbb{P}^1$ are two distinct birational minimal surfaces.

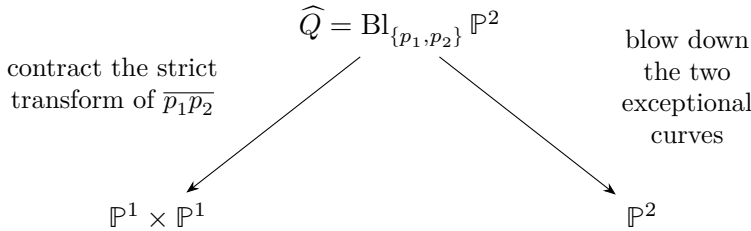


Figure 2.1: Two distinct minimal surfaces.

From now on, until the characteristic-zero convention in Section 2.3A, varieties are over an arbitrary algebraically closed field k . We follow [Deb01, 1.10]. A rational curve means an integral projective curve whose normalization is $\mathbb{P}_k^1$.

Definition 2.1.1. Let $\pi : Y \rightarrow X$ be a birational morphism of varieties. The exceptional locus $\text{Ex}(\pi)$ is the locus of points of Y where π is not a local isomorphism. We denote it by E .

The set E is closed, since the local-isomorphism locus is open. The rational inverse π^{-1} has its domain of definition in X , not in Y .

Assume now that X is normal and π is proper. Then π is surjective and has connected fibers, by Zariski's connectedness theorem [GW23, 24.58]. A zero-dimensional fiber is a single point; properness and the quasi-finite locus reduce a neighborhood to a finite birational morphism, which is an isomorphism because X is normal. Thus a fiber meets E exactly when it is positive-dimensional. Connectedness then implies that the entire fiber lies in E : a point where π is a local isomorphism would be isolated in that fiber. Consequently,

$$E = \pi^{-1}(\pi(E)), \quad \pi(E) = \{x \in X \mid \dim \pi^{-1}(x) \geq 1\}.$$

The latter is closed in X , by upper semicontinuity of fiber dimension for a proper morphism. If it had a codimension-one component W , then

$$\dim \pi^{-1}(W) \geq \dim W + 1 = \dim Y.$$

This would force the proper closed subset $\pi^{-1}(W)$ to equal the irreducible variety Y , a contradiction. Hence $\operatorname{codim}_X \pi(E) \geq 2$. The largest open set on which the inverse is defined is $X \setminus \pi(E)$.

Remark 2.1.2 (Proper and birational is surjective). Since π is proper, $\pi(Y)$ is closed. Since it is birational, this image contains a dense open subset of X . Thus $\pi(Y) = X$.

Proposition 2.1.3 (Van der Waerden). *Let $\pi : Y \rightarrow X$ be a birational morphism of varieties. If X is normal and $\mathbb{Q}$ -factorial, then every irreducible component of $\operatorname{Ex}(\pi)$ has codimension one in Y .*

Proof. Let y be a closed point of E and $x = \pi(y)$. Identify the function fields using π^* . The local inclusion $\mathcal{O}_{X,x} \subset \mathcal{O}_{Y,y}$ is strict; otherwise the morphism would be a local isomorphism at y . As both residue fields are k , we can choose $t \in \mathfrak{m}_y \setminus \mathcal{O}_{X,x}$ by subtracting its residue from an element outside $\mathcal{O}_{X,x}$.

Write $\operatorname{div}_X(t) = D' - D''$ with effective Weil divisors having no common components. By $\mathbb{Q}$ -factoriality, some mD' and mD'' are Cartier. Their local equations can be chosen so that

$$t^m = u/v, \quad u, v \in \mathcal{O}_{X,x}.$$

Here $v \in \mathfrak{m}_x$, for otherwise $t^m \in \mathcal{O}_{X,x}$ and normality would imply $t \in \mathcal{O}_{X,x}$. Also $u = t^m v \in \mathfrak{m}_y \cap \mathcal{O}_{X,x} = \mathfrak{m}_x$. The subscheme $Z = (u = v = 0)$ has codimension two near x , since the two divisors have no common components. On a neighborhood of y where t is regular, its inverse image is defined by

$$t^m v = v = 0,$$

hence by the single nonzero equation $v = 0$. Every irreducible component through y of this local zero locus has codimension one. Such a component is exceptional: outside E , a local isomorphism cannot map a divisor into the codimension-two set Z . Thus every closed point of E lies on a codimension-one component of E . This proves the assertion. $\square$

Ramification divisor of a morphism between smooth varieties

Assume X and Y are smooth of the same dimension n . The map

$$\pi^*\Omega_X \longrightarrow \Omega_Y$$

induces $\pi^*\omega_X \rightarrow \omega_Y$, hence a section of

$$\pi^*\omega_X^{-1} \otimes \omega_Y \cong \mathcal{O}_Y(K_Y - \pi^*K_X).$$

If π is dominant and generically separable, this section is nonzero. Its zero divisor is the effective ramification divisor $\text{Ram}(\pi)$, locally cut out by the Jacobian determinant. Thus, choosing compatible canonical divisors,

$$K_Y = \pi^*K_X + \text{Ram}(\pi).$$

For a birational morphism, separability is automatic and $\text{Supp } \text{Ram}(\pi) = \text{Ex}(\pi)$. The multiplicities in $\text{Ram}(\pi)$ need not be one.

Proposition 2.1.4 (Abhyankar's lemma). *Let X and Y be varieties with X smooth, and let $\pi : Y \rightarrow X$ be a proper birational morphism which is not an isomorphism. Through a general point of each component of $\text{Ex}(\pi)$, there is a rational curve contracted by π .*

Proof. Replace Y by its normalization. This is finite over Y , and a nonconstant image of a rational curve is rational by Lüroth's theorem. Thus it suffices to work on the normalization and descend the curves. By Proposition 2.1.3, the exceptional components are divisors. Fix one such component E with generic point η_E ; Y is smooth near η_E .

We construct successive blow-ups of the centers of E , working on neighborhoods of η_E . Let $X_0 = X$. As long as the center Z_{i-1} of E on X_{i-1} has codimension at least two, take an open neighborhood U_{i-1} of its generic point on which the center is smooth, and blow it up:

$$\varepsilon_i : X_i = \text{Bl}_{Z_{i-1} \cap U_{i-1}} U_{i-1} \longrightarrow U_{i-1}.$$

The ideal of the center pulls back to a nonzero principal ideal in the DVR $\mathcal{O}_{Y, \eta_E}$. After shrinking a neighborhood V_i of η_E in Y , it is invertible there. The universal property of the blow-up gives a birational morphism $\pi_i : V_i \rightarrow X_i$ and the diagram

$$\begin{array}{ccccc} V_i & \xrightarrow{\pi_i} & X_i & \xrightarrow{\varepsilon_i} & U_{i-1} \\ & & & & \subseteq X_{i-1}. \end{array}$$

This is the precise reason that the lift exists; smoothness of E alone would not make the pulled-back ideal invertible everywhere.

Let E_i be the exceptional divisor of ε_i and

$$c_i = \text{codim}_{U_{i-1}}(Z_{i-1} \cap U_{i-1}) - 1 > 0.$$

The blow-up formula gives

$$K_{X_i} = \varepsilon_i^* K_{U_{i-1}} + c_i E_i.$$

Moreover, $\pi_i^* E_i$ has positive integral multiplicity along E . Since the remaining ramification divisor on the smooth neighborhood V_i is effective, repeated pullback gives, at η_E ,

$$K_Y - \pi^* K_X \geq (c_1 + \cdots + c_i)E.$$

More explicitly, each summand is c_j times the pullback of E_j , whose E -multiplicity is at least one. The fixed integer

$$a = \text{ord}_E(K_Y - \pi^* K_X)$$

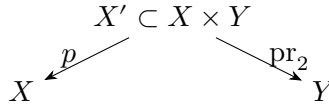
therefore satisfies $a \geq c_1 + \cdots + c_i \geq i$. The process must terminate.

At the last step, the center of E is a divisor on the smooth X_i , and it is the exceptional divisor E_i of the last blow-up. A birational morphism to a normal variety is an isomorphism at the generic point of a divisor that it dominates. Thus E is birational to E_i , which is a projective-space bundle over the smooth center of ε_i .

Through a general point of E_i choose a line in its projective-space fiber. The birational inverse to E gives a rational map from that line to Y , defined near the chosen point. Its composite with π is constant. Since $Y \rightarrow X$ is proper, the valuative criterion extends this map over the whole $\mathbb{P}^1$. Its image is a nonconstant rational curve through the chosen general point of E , contracted by π . Here we use the original proper morphism; shrinking Y did not make the resulting open neighborhood proper. $\square$

Corollary 2.1.5. *Let X be smooth and Y proper, with no rational curves. Every rational map $f : X \dashrightarrow Y$ is defined everywhere.*

Proof. Let $X' \subset X \times Y$ be the closure of the graph. The first projection $p : X' \rightarrow X$ is proper and birational. If it is not an isomorphism, Proposition 2.1.4 supplies a rational curve contracted by p . The second projection also contracts it because Y contains no rational curves. But a curve in $X \times Y$ cannot be contracted by both projections. Thus $\text{Ex}(p)$ is empty, p is an isomorphism, and f is a morphism.



An exceptional rational curve would map
to a point under both projections.

Figure 2.2: The two projections cannot both contract a curve in the graph.

$\square$

Remark 2.1.6. The valuation-theoretic form of the termination argument is discussed in [KM98, 2.45] and the appendix of [Kol96]. In the smooth-target situation above, the fixed ramification coefficient at η_E supplies the required termination bound. Thus the proof must include this step; it is not enough simply to repeat the blow-ups indefinitely.

Proposition 2.1.7 (Negativity lemma in its simplest form). *Let X and Y be smooth projective varieties and $\pi : Y \rightarrow X$ a birational morphism which is not an isomorphism. There is a rational curve C on Y , contracted by π , such that $(K_Y \cdot C) < 0$.*

Proof. Put $E = \text{Ex}(\pi)$ and $B = \pi(E)$, so $\text{codim}_X B \geq 2$. Choose a closed point $x \in B$. Using sufficiently high very ample multiples, take $\dim X - 2$ general hypersurface sections through x to cut out a smooth projective surface S such that $S \cap B$ is finite. Smoothness at the fixed point is justified by Lemma 2.1.8; proper intersection with B follows by general choice. When $\dim X = 2$, take $S = X$.

On S , choose two smooth irreducible members C_0, C of the same sufficiently ample linear system, with $C_0 \cap B = \{x\}$ and $C \cap B = \emptyset$. Hence

$$(K_X \cdot C_0) = (K_X \cdot C).$$

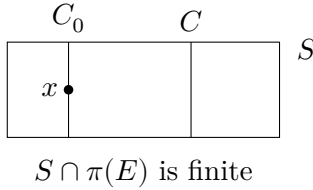


Figure 2.3: Construction of the two smooth curves C_0 and C .

Let $C' = \pi^{-1}(C)$ and let C'_0 be the strict transform of C_0 . Since C' avoids E ,

$$(K_Y \cdot C') = (\pi^* K_X + \text{Ram}(\pi)) \cdot C' = (K_X \cdot C).$$

The curve C'_0 meets E : its proper map to C_0 is surjective, and the fiber above x lies in E . It is not contained in E , so

$$\begin{aligned} (K_Y \cdot C'_0) &> (\pi^* K_X \cdot C'_0) = (K_X \cdot C_0) \\ &= (K_X \cdot C) = (K_Y \cdot C'). \end{aligned}$$

Resolve the rational map $\pi^{-1}|_S : S \dashrightarrow Y$ by point blow-ups:

$$g : \tilde{S} \xrightarrow{\varepsilon} S \dashrightarrow Y, \quad \pi \circ g = \iota_S \circ \varepsilon.$$

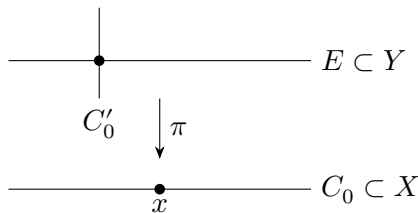


Figure 2.4: The strict transform C'_0 meets the exceptional locus.

$$\begin{array}{ccc} \tilde{S} & \xrightarrow{g} & Y \\ \varepsilon \downarrow & & \downarrow \pi \\ S & \hookrightarrow & X. \end{array}$$

Figure 2.5: Blowing up points, including possibly infinitely near points, to resolve the map.

All centers lie over the finite set $S \cap B$. This resolves the map from the surface; it does not assert a factorization of the original higher-dimensional morphism π .

Write $C'' = \varepsilon^*C$ and let C''_0 be the strict transform of C_0 . Then $g_*C'' = C'$ and $g_*C''_0 = C'_0$, while

$$\varepsilon^*C_0 = C''_0 + \sum_i m_i E_i, \quad m_i \geq 0.$$

Here the E_i are exceptional prime curves on the final surface. They are smooth rational curves, but need not all still have self-intersection -1 ; the m_i are their total-transform multiplicities. Since $C \sim_{\text{lin}} C_0$ on S , pushing the relation forward gives an equality modulo rational equivalence of 1-cycles on Y :

$$C' \sim_{\text{rat}} C'_0 + \sum_i m_i g_* E_i.$$

In particular,

$$(K_Y \cdot C') = (K_Y \cdot C'_0) + \sum_i m_i (K_Y \cdot g_* E_i).$$

The strict inequality above implies that some $(K_Y \cdot g_* E_i) < 0$. Its image is therefore a curve, rational by Lüroth's theorem, and it is contracted by π because $\varepsilon(E_i)$ is a point. The mapping degree in $g_* E_i$ is positive, so $K_Y \cdot g(E_i) < 0$ as required. □

When taking a first look at this argument, I thought the fixed-point Bertini step was trivial, but my advisor had doubts about it. We therefore give a complete proof here. The following formulation records exactly the sufficiently ample version used above.

$$E_i \cong \mathbb{P}^1 \xrightarrow{g} g(E_i) \subset E \subset Y \xrightarrow{\pi} \{x_i\} \subset X.$$

Figure 2.6: A noncontracted image $g(E_i)$ with negative canonical degree.

Lemma 2.1.8. *Let X be a smooth projective variety of dimension at least two, $x \in X$ a closed point, and L a very ample divisor. For $m \gg 0$, a general member of $|mL|$ passing through x is smooth and irreducible. Thus it is a hyperplane section for the embedding given by $|mL|$.*

Proof. Let $\pi : \widetilde{X} = \text{Bl}_x X \rightarrow X$, with $E \cong \mathbb{P}^{n-1}$. For $m \gg 0$, $m\pi^*L - E$ is very ample. By Bertini, a general

$$H' \in |m\pi^*L - E|$$

is smooth and irreducible, and does not contain E . Its image H is smooth away from x . The pullback line bundle $\pi^*\mathcal{O}_X(mL)|_E$ is trivial, since E maps to a point, whereas $\mathcal{O}_{\widetilde{X}}(-E)|_E \cong \mathcal{O}_E(1)$. Therefore

$$\mathcal{O}_{\widetilde{X}}(H')|_E \cong \mathcal{O}_E(1).$$

Locally at x , the equation f of H has a nonzero linear initial form F , because its total transform has multiplicity exactly one along E . Thus

$$f = F + \text{terms of order at least two},$$

so H is smooth at x . The identity $\pi_*\mathcal{O}_{\widetilde{X}}(-E) = \mathfrak{m}_x$ identifies this linear system with the members of $|mL|$ through x , proving the general-member assertion. $\square$

2.1B Parametrizing Rational Curves

Let k be algebraically closed, not necessarily of characteristic zero. For a projective variety $X \subseteq \mathbb{P}_k^N$, we aim to construct a scheme $\text{Mor}(\mathbb{P}_k^1, X)$ parametrizing morphisms $\mathbb{P}_k^1 \rightarrow X$. This gives a precise meaning to deforming a morphism.

To begin with, take $X = \mathbb{P}_k^N$, with $N \geq 1$. A morphism $f : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^N$ is given by homogeneous polynomials $F_0, \dots, F_N$ of the same degree $d \geq 0$:

$$f([u : v]) = [F_0(u, v) : \dots : F_N(u, v)],$$

with no common zero in $\mathbb{P}_k^1$. Two such tuples of the same degree define the same morphism precisely when they differ by one common nonzero scalar. The scalar cannot be chosen independently for each coordinate.

Each F_i has $d + 1$ coefficients, so the tuples up to simultaneous scaling form the projective space

$$\mathbb{P}(k^{(d+1)(N+1)}) \cong \mathbb{P}_k^{(d+1)(N+1)-1}.$$

Inside it, let $\text{Mor}_d(\mathbb{P}_k^1, \mathbb{P}_k^N)$ consist of tuples with no common zero. The degree here is $\deg f^* \mathcal{O}_{\mathbb{P}^N}(1)$; it includes the mapping degree onto the image.

Lemma 2.1.9. *The subset $\text{Mor}_d(\mathbb{P}_k^1, \mathbb{P}_k^N) \subset \mathbb{P}_k^{(d+1)(N+1)-1}$ is Zariski open, hence a quasi-projective variety.*

Proof. If $d = 0$, every nonzero tuple of constants defines a constant morphism, and this subset is all of $\mathbb{P}_k^N$. Suppose $d > 0$, and put $I = (F_0, \dots, F_N) \subset k[U, V]$. The absence of a projective common zero is equivalent, by the Nullstellensatz, to

$$\sqrt{I} = (U, V), \quad \text{equivalently} \quad (U, V)^m \subset I \text{ for some } m \geq d.$$

Equivalently, for some $m \geq d$, the linear map

$$\begin{aligned} \varphi_m : (k[U, V]_{m-d})^{N+1} &\longrightarrow k[U, V]_m, \\ (G_0, \dots, G_N) &\longmapsto \sum_i G_i F_i \end{aligned}$$

is surjective. Its matrix entries are linear in the coefficients of the F_i , and surjectivity means that at least one $(m+1)$ -minor is nonzero. This is open for each m . Taking the union of these open sets proves the assertion. Equivalently, the bad locus is the intersection, over all m , of the closed rank-deficiency loci. $\square$

Thus all morphisms are parametrized by the locally noetherian disjoint union

$$\text{Mor}(\mathbb{P}_k^1, \mathbb{P}_k^N) = \coprod_{d \geq 0} \text{Mor}_d(\mathbb{P}_k^1, \mathbb{P}_k^N).$$

Remark 2.1.10. The degree- d morphisms fit into a universal evaluation morphism

$$\begin{aligned} f^{\text{univ}} : \mathbb{P}_k^1 \times \text{Mor}_d(\mathbb{P}_k^1, \mathbb{P}_k^N) &\longrightarrow \mathbb{P}_k^N, \\ ([u : v], [F_0 : \dots : F_N]) &\longmapsto [F_0(u, v) : \dots : F_N(u, v)]. \end{aligned}$$

If $X \subseteq \mathbb{P}_k^N$ is a closed subscheme defined by homogeneous equations $G_1, \dots, G_a$, the condition that the image lies in X is

$$G_j(F_0(U, V), \dots, F_N(U, V)) = 0 \quad (1 \leq j \leq a).$$

Each coefficient of these polynomials must vanish. These equations define the closed subscheme $\text{Mor}_d(\mathbb{P}_k^1, X)$ of the open scheme above, and

$$\text{Mor}(\mathbb{P}_k^1, X) = \coprod_{d \geq 0} \text{Mor}_d(\mathbb{P}_k^1, X).$$

These are quasi-projective schemes; they need not be reduced or irreducible, even when X is a variety.

2.1C Parametrizing Morphisms

We admit the following theorem generalizing the preceding construction.

Theorem 2.1.11 (Grothendieck's Hilbert scheme theorem). *Let X be a quasi-projective k -scheme and Y a projective k -scheme. There is a scheme $\text{Mor}(Y, X)$ locally of finite type over k , parametrizing morphisms $Y \rightarrow X$ in the following sense:*

(a) *There is a universal evaluation morphism*

$$f^{\text{univ}} : Y \times_k \text{Mor}(Y, X) \longrightarrow X.$$

(b) *For every k -scheme T and morphism $f : Y \times_k T \rightarrow X$, there is a unique morphism $\varphi : T \rightarrow \text{Mor}(Y, X)$ such that*

$$f = f^{\text{univ}} \circ (\text{id}_Y \times \varphi).$$

The universal property is expressed by the commutative diagram

$$\begin{array}{ccc} Y \times T & & \\ \text{id}_Y \times \varphi \downarrow & \searrow f & \\ Y \times \text{Mor}(Y, X) & \xrightarrow{f^{\text{univ}}} & X \end{array}$$

The theorem applies in particular to varieties, and also to the finite and nonreduced schemes used below. The space is constructed by parametrizing graphs in $Y \times X$.

Fix very ample line bundles A on Y and L on X , and use the product polarization $A \boxtimes L$. The Hilbert polynomial to fix is the polynomial of the graph:

$$P_f(m) = \chi(Y, A^{\otimes m} \otimes f^* L^{\otimes m}).$$

The locus $\text{Mor}_P(Y, X)$ with this polynomial is quasi-projective and of finite type, and

$$\text{Mor}(Y, X) = \coprod_P \text{Mor}_P(Y, X).$$

There are only countably many such polynomials. For a curve Y , fixing this polynomial is equivalent to fixing $\deg f^* L$, since A and Y are fixed. In higher dimension, one must retain the $A^{\otimes m}$ factor in this boundedness statement.

Remark 2.1.12. Projectivity of the source is essential here. For example, infinitesimal deformations of the zero map $\mathbb{A}_k^1 \rightarrow \mathbb{A}_k^N$ form $k[t]^{\oplus N}$, which is infinite-dimensional. Thus this functor cannot be represented by a scheme locally of finite type over k .

In moduli language, the contravariant functor

$$\mathcal{M}or(Y, X) : \text{Sch}_k^{\text{op}} \rightarrow \text{Sets}, \quad T \mapsto \text{Hom}_k(Y \times_k T, X)$$

is represented by $\text{Mor}(Y, X)$. It is naturally isomorphic to the functor

$$T \mapsto \text{Hom}_k(T, \text{Mor}(Y, X)).$$

Given $\varphi : T \rightarrow \text{Mor}(Y, X)$, its evaluation is

$$\text{ev} : Y \times T \rightarrow X, \quad (y, t) \mapsto \varphi(t)(y).$$

This is a fine moduli problem for morphisms with the source and target fixed; we do not quotient by automorphisms of the source.

Example 2.1.13. If $Y = \text{Spec } k$, then $\text{Mor}(Y, X) \cong X$ as schemes. Its k -points are $\text{Hom}_k(\text{Spec } k, X) = X(k)$, rather than the scheme X itself. The universal evaluation is the canonical isomorphism $\text{Spec } k \times X \rightarrow X$. More generally, if Y is a reduced projective zero-dimensional k -scheme, then Y is a finite disjoint union of k -points and

$$\text{Mor}(Y, X) \cong X^{|Y|}.$$

Example 2.1.14 (Dual numbers). For $Y = \text{Spec } k[\epsilon]/(\epsilon^2)$, a k -morphism $Y \rightarrow X$ consists of a point $x \in X(k)$ and a tangent vector in $T_{X,x}$. If X is smooth, $\text{Mor}(Y, X)$ is the total space of the tangent bundle T_X ; see [GW20, (6.4)].

Example 2.1.15. We use the universal property to determine $T_{\text{Mor}(Y, X), [f]}$. Its elements are morphisms

$$\varphi : \text{Spec } k[\epsilon]/(\epsilon^2) \rightarrow \text{Mor}(Y, X)$$

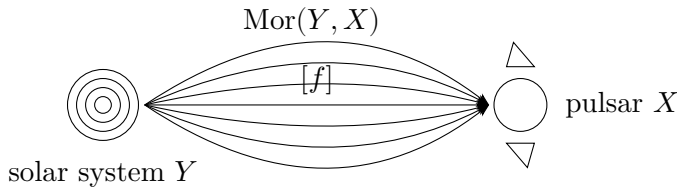
whose restriction to $\text{Spec } k$ is $[f]$. They correspond to extensions

$$f_\epsilon : Y \times \text{Spec } k[\epsilon]/(\epsilon^2) \rightarrow X, \quad f_\epsilon|_{Y \times \text{Spec } k} = f.$$

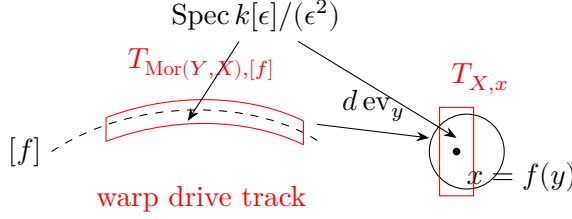
These are first-order deformations of the *morphism* f , with Y fixed. The diagram is

$$\begin{array}{ccc} Y \times \text{Spec } k[\epsilon]/(\epsilon^2) & \xrightarrow{f_\epsilon} & X \\ \text{id} \times \varphi \downarrow & & \parallel \\ Y \times \text{Mor}(Y, X) & \xrightarrow{f^{\text{univ}}} & X. \end{array}$$

We can view this as a family of tracks from a fixed source to the target. The “solar system / pulsar” picture illustrates the parameter space of maps:



A tangent vector at the parameter point $[f]$ is an infinitesimal change of the track. At a chosen $y \in Y$, it induces the infinitesimal motion of the landing point $x = f(y)$:



Proposition 2.1.16 (Tangent space of $\text{Mor}(Y, X)$). *For Y projective, X quasi-projective, and a morphism $f : Y \rightarrow X$, there is a canonical isomorphism*

$$T_{\text{Mor}(Y,X),[f]} \cong H^0(Y, \mathcal{H}om(f^*\Omega_X, \mathcal{O}_Y)).$$

Proof. First compute locally on affine schemes $\text{Spec } B$ and $\text{Spec } A$. This is a calculation of deformations of ring maps; it does not require a representing scheme for morphisms between arbitrary affine schemes. If $f^\# : A \rightarrow B$ is the original map, a first-order lift has the form

$$f_\epsilon^\#(a) = f^\#(a) + \epsilon g(a).$$

It is a k -algebra homomorphism precisely when g is k -linear and

$$\begin{aligned} f_\epsilon^\#(aa') &= f^\#(aa') + \epsilon g(aa') \\ &= f^\#(a)f^\#(a') + \epsilon(f^\#(a)g(a') + f^\#(a')g(a)), \end{aligned}$$

that is, $g(aa') = f^\#(a)g(a') + f^\#(a')g(a)$. By the universal property of differentials,

$$\text{Der}_k(A, B) \cong \text{Hom}_A(\Omega_{A/k}, B) \cong \text{Hom}_B(\Omega_{A/k} \otimes_A B, B).$$

Here B is an A -module via $f^\#$.

In general, take affine opens $V_i \subset Y$, $U_i \subset X$ with $f(V_i) \subset U_i$. The underlying topological spaces do not change under the dual-number thickening, so a lift restricts to $V_i \times \text{Spec } k[\epsilon]/(\epsilon^2) \rightarrow U_i$. The derivations just computed must agree on overlaps. They therefore glue to a global section of $\mathcal{H}om(f^*\Omega_X, \mathcal{O}_Y)$, and every such section gives a lift. This proves the assertion. $\square$

Corollary 2.1.17 (A very important result). *If X is smooth along $f(Y)$, then*

$$T_{\text{Mor}(Y,X),[f]} \cong H^0(Y, f^*T_X).$$

Proof. The sheaf Ω_X is locally free on a neighborhood of $f(Y)$, hence

$$\mathcal{H}om(f^*\Omega_X, \mathcal{O}_Y) \cong (f^*\Omega_X)^\vee \cong f^*T_X.$$

$\square$

Example 2.1.18. For smooth projective X , $\text{Aut}(X)$ is an open subscheme of $\text{Mor}(X, X)$. At an automorphism f , its tangent space is $H^0(X, f^*T_X) \cong H^0(X, T_X)$. If $H^0(X, T_X) = 0$, the automorphism scheme is zero-dimensional and reduced, hence discrete over k . It need not be finite: local finite type does not imply finite type for the entire automorphism scheme.

We now discuss the local structure of the parameter space.

Lemma 2.1.19. *Let $(R, \mathfrak{m})$ be a noetherian local k -algebra, essentially of finite type, with residue field $R/\mathfrak{m} = k$. Let $I \subset \mathfrak{m}$ satisfy $\mathfrak{m}I = 0$. Let Y be projective and $f : Y \rightarrow X$ a morphism, with X smooth along $f(Y)$. For an extension*

$$f_{R/I} : Y \times \text{Spec}(R/I) \rightarrow X,$$

*there is an obstruction class in $H^1(Y, f^*T_X) \otimes_k I$ whose vanishing is equivalent to the existence of a lift $f_R : Y \times \text{Spec } R \rightarrow X$.*

Proof. Since $\mathfrak{m}I = 0$, we have $I^2 = 0$, and I is a finite-dimensional k -vector space. The source Y_R is flat over R , so its ideal for $Y_{R/I} \subset Y_R$ is $\mathcal{O}_Y \otimes_k I$, supported on the closed fiber.

The given map has image in the smooth locus of X . Indeed, the inverse image of its complement is closed in the proper R/I -scheme $Y_{R/I}$; its image in $\text{Spec}(R/I)$ is closed and misses the closed point, so it is empty. Cover Y_R by affine opens whose reductions map into smooth affine opens of X . On each such open, formal smoothness supplies a lift. In ring notation, this is the lifting across the square-zero surjection

$$B_R \twoheadrightarrow B_R/IB_R, \quad IB_R \cong (B_R/\mathfrak{m}B_R) \otimes_k I.$$

The affine lifting diagram is

$$\begin{array}{ccc} & & B_R \\ & \nearrow \text{dashed} & \downarrow \\ A & \xrightarrow{\quad} & B_R/IB_R \end{array}$$

where the diagonal map is the sought lift. For a smooth affine target $\text{Spec } A$, two lifts differ by a derivation. On the closed fiber, writing $B = B_R/\mathfrak{m}B_R$, the space of differences is

$$\begin{aligned} \text{Hom}_A(\Omega_{A/k}, B \otimes_k I) &\cong \text{Hom}_B(\Omega_{A/k} \otimes_A B, B \otimes_k I) \\ &\cong \text{Hom}_B(\Omega_{A/k} \otimes_A B, B) \otimes_k I. \end{aligned}$$

This is the local section space of $f^*T_X \otimes_k I$.

On overlaps, the differences of local lifts form a Čech 1-cocycle. The cocycle condition follows by adding the differences of three lifts. Since the source is separated, affine intersections compute the cohomology of this quasi-coherent sheaf. The resulting class lies in $H^1(Y, f^*T_X) \otimes_k I$. If it vanishes, subtracting a 0-cochain from the local lifts makes them agree, giving a global lift. Conversely, a global lift makes this cocycle a coboundary.

When R is Artinian the usual product affine cover of Y suffices. The affine-cover argument on Y_R also handles the essentially finite type local rings used in Theorem 2.1.23; their spectra are not necessarily nilpotent thickenings of a single point. $\square$

Remark 2.1.20. Choose a k -basis of I , of dimension s . Then

$$\begin{aligned} \mathrm{Hom}_B(\Omega_A \otimes_A B, B \otimes_k I) &\cong \mathrm{Hom}_B(\Omega_A \otimes_A B, B^{\oplus s}) \\ &\cong \mathrm{Hom}_B(\Omega_A \otimes_A B, B)^{\oplus s} \\ &\cong \mathrm{Hom}_B(\Omega_A \otimes_A B, B) \otimes_k I. \end{aligned}$$

One must choose a basis, not just an arbitrary list of ideal generators and identify its length with $\dim_k I$.

Remark 2.1.21 (An important case). Take $R = k[\epsilon]/(\epsilon^m)$, $I = (\epsilon^{m-1})$, $\mathfrak{m} = (\epsilon)$, with $m \geq 2$. The obstruction to extending a deformation from $Y \times \mathrm{Spec} k[\epsilon]/(\epsilon^{m-1})$ to $Y \times \mathrm{Spec} k[\epsilon]/(\epsilon^m)$ lies in $H^1(Y, f^*T_X)$, using the basis ϵ^{m-1} of I .

Lemma 2.1.22. *Let $(R, \mathfrak{m})$ be a noetherian local ring and $I \subset \mathfrak{m}^2$ an ideal. If the projection $\pi : R \rightarrow R/I$ admits a ring section $s : R/I \rightarrow R$, then $I = 0$.*

Proof. For $a, b \in \mathfrak{m}$, write $s(\pi(a)) = a + a'$, $s(\pi(b)) = b + b'$ with $a', b' \in I$. Then

$$s(\pi(ab)) = (a + a')(b + b') \in ab + \mathfrak{m}I.$$

Every element of $\mathfrak{m}^2$ is a finite sum of such products, so for $x \in I \subset \mathfrak{m}^2$,

$$0 = s(\pi(x)) \in x + \mathfrak{m}I.$$

Thus $I = \mathfrak{m}I$, and Nakayama's lemma gives $I = 0$. $\square$

Theorem 2.1.23 (Local structure of $\mathrm{Mor}(Y, X)$). *Let X, Y be projective varieties and $f : Y \rightarrow X$ a morphism with X smooth along $f(Y)$. Near $[f]$, the scheme $\mathrm{Mor}(Y, X)$ can be defined by at most $h^1(Y, f^*T_X)$ equations in a smooth variety of dimension $h^0(Y, f^*T_X)$.*

Proof. Because $\mathrm{Mor}(Y, X)$ is locally of finite type, an affine neighborhood of $[f]$ is cut out by finitely many polynomials $P_1, \dots, P_m$ in $\mathbb{A}^n$. The rank of their Jacobian at $[f]$ is exactly

$$r = n - \dim T_{\mathrm{Mor}(Y, X), [f]} = n - h^0(Y, f^*T_X).$$

Choose r of the equations whose differentials are independent. They cut out a variety V , smooth near $[f]$, of dimension $h^0(Y, f^*T_X)$. After shrinking, $\text{Mor}(Y, X)$ is a closed subscheme of V and their tangent spaces at $[f]$ coincide.

Set $R = \mathcal{O}_{V, [f]}$, with maximal ideal $\mathfrak{m}$, and let I define the local ring of $\text{Mor}(Y, X)$. Equality of tangent spaces gives $I \subset \mathfrak{m}^2$. By Nakayama's lemma, it suffices to show

$$\dim_k I/\mathfrak{m}I \leq h^1(Y, f^*T_X).$$

The natural map $\text{Spec } R/I \rightarrow \text{Mor}(Y, X)$ gives $f_{R/I}$. Lemma 2.1.19, applied to the local ring $R/\mathfrak{m}I$ and its small ideal $I/\mathfrak{m}I$, gives an obstruction

$$o = \sum_{i=1}^{h^1} a_i \otimes \bar{b}_i \in H^1(Y, f^*T_X) \otimes_k I/\mathfrak{m}I,$$

where the a_i form a basis and $b_i \in I$. Put

$$J = \mathfrak{m}I + (b_1, \dots, b_{h^1}).$$

The image of o in $H^1(Y, f^*T_X) \otimes_k I/J$ is zero, so the map lifts to $Y \times \text{Spec } R/J \rightarrow X$.

By the universal property, this gives $\text{Spec } R/J \rightarrow \text{Mor}(Y, X)$. Since R/J is local and its closed point maps to $[f]$, this morphism factors through the local scheme $\text{Spec } \mathcal{O}_{\text{Mor}(Y, X), [f]} = \text{Spec } R/I$. This is the universal property of localization; $\text{Spec } R/I \rightarrow \text{Mor}(Y, X)$ need not be an open immersion. Its restriction to $\text{Spec } R/I$ is the original map, so we obtain a section of

$$R/J \twoheadrightarrow R/I.$$

Equivalently, the diagram on local schemes is

$$\begin{array}{ccc} \text{Spec } R/J & \xrightarrow{\quad} & \text{Spec } R/I \\ \uparrow & \nearrow \text{id} & \\ \text{Spec } R/I & & \end{array}$$

The kernel I/J lies in the square of the maximal ideal of R/J . Lemma 2.1.22 therefore gives $I = J$. Thus the images of the b_i generate $I/\mathfrak{m}I$, as required. Shrinking the neighborhood realizes these local generators as at most h^1 defining equations. $\square$

Corollary 2.1.24. *Every irreducible component of $\text{Mor}(Y, X)$ through $[f]$ has dimension at least*

$$h^0(Y, f^*T_X) - h^1(Y, f^*T_X).$$

This follows from the dimension bound for a scheme cut out by a given number of equations in a smooth variety. It is the classical conclusion of [Kol96, I, 2.16], over an arbitrary algebraically closed field.

2.1D Morphisms with Fixed Points

Fix a closed subscheme $B \subseteq Y$ and a morphism $g : B \rightarrow X$. Consider restriction

$$\rho : \text{Mor}(Y, X) \rightarrow \text{Mor}(B, X), \quad f \mapsto f|_B,$$

and denote its scheme-theoretic fiber over $[g]$ by $\text{Mor}(Y, X; g)$. Here Theorem 2.1.11 is used for the projective scheme B , which need not be reduced.

If $f|_B = g$ and X is smooth along $f(Y)$, the tangent map is restriction

$$d\rho : H^0(Y, f^*T_X) \longrightarrow H^0(B, g^*T_X).$$

Tensoring $0 \rightarrow \mathcal{I}_B \rightarrow \mathcal{O}_Y \rightarrow \mathcal{O}_B \rightarrow 0$ with the locally free f^*T_X and taking cohomology gives

$$\begin{aligned} 0 \rightarrow H^0(Y, f^*T_X \otimes \mathcal{I}_B) &\rightarrow H^0(Y, f^*T_X) \xrightarrow{d\rho} H^0(B, g^*T_X) \\ &\rightarrow H^1(Y, f^*T_X \otimes \mathcal{I}_B) \rightarrow \cdots \end{aligned}$$

Thus $T_{\text{Mor}(Y, X; g), [f]} = H^0(Y, f^*T_X \otimes \mathcal{I}_B)$.

Theorem 2.1.25 (Local structure with fixed points). *With X, Y, B, g, f as above, $\text{Mor}(Y, X; g)$ is locally defined near $[f]$ by at most $h^1(Y, f^*T_X \otimes \mathcal{I}_B)$ equations in a smooth variety of dimension $h^0(Y, f^*T_X \otimes \mathcal{I}_B)$.*

The obstruction proof is the same as above, with differences of lifts required to vanish along B . Local lifts agreeing with the prescribed map on B exist by formal smoothness: combine the maps on $Y_{R/I}$ and B_R , which agree on $B_{R/I}$, and lift across their square-zero union in Y_R . The sheaf of differences is $f^*T_X \otimes \mathcal{I}_B \otimes_k I$. The cocycle argument and Theorem 2.1.23 then apply.

2.1E Morphisms from a Curve

Let C be a smooth projective curve and $f : C \rightarrow X$ a morphism, with X smooth along $f(C)$. By Corollary 2.1.24,

$$\dim_{[f]} \text{Mor}(C, X) \geq h^0(C, f^*T_X) - h^1(C, f^*T_X) = \chi(C, f^*T_X).$$

Writing $n = \dim X$, the Riemann–Roch calculation is

$$\text{ch}(f^*T_X) = n + c_1(f^*T_X), \quad \text{td}(T_C) = 1 + \frac{1}{2}c_1(T_C).$$

Taking the degree-one part of the product,

$$\begin{aligned} \chi(C, f^*T_X) &= \deg(\text{ch}(f^*T_X) \text{td}(T_C))_1 \\ &= \deg c_1(f^*T_X) + \frac{n}{2} \deg c_1(T_C) \\ &= \deg f^*\omega_X^{-1} + \frac{n}{2} \deg(-K_C) \\ &= -(K_X \cdot f_*C) + n(1 - g(C)). \end{aligned}$$

Here $c_1(f^*T_X) = c_1(\det f^*T_X)$ and f_*C is the pushforward cycle, including the mapping degree. Thus

$$\dim_{[f]} \text{Mor}(C, X) \geq -(K_X \cdot f_*C) + (1 - g(C)) \dim X.$$

For a finite subscheme $B \subset C$, the same exact sequence gives

$$\chi(C, f^*T_X \otimes \mathcal{I}_B) = \chi(C, f^*T_X) - \chi(B, f^*T_X|_B).$$

Since B is zero-dimensional,

$$\chi(B, f^*T_X|_B) = h^0(B, f^*T_X|_B) = \text{length}(B) \dim X.$$

By Theorem 2.1.25,

$$\begin{aligned} \dim_{[f]} \text{Mor}(C, X; f|_B) &\geq \chi(C, f^*T_X \otimes \mathcal{I}_B) \\ &= -(K_X \cdot f_*C) + (1 - g(C) - \text{length}(B)) \dim X. \end{aligned}$$

In particular, for a single point $c_0 \in C$,

$$\dim_{[f]} \text{Mor}(C, X; f|_{\{c_0\}}) \geq -(K_X \cdot f_*C) - g(C) \dim X.$$

This is the dimension estimate used in [KM98, pp. 12–13] and in the bend-and-break arguments below.

2.2 “Bend and Break” Lemmas

Olivier Debarre writes in [Deb01, Chapter 3]:

The whole story began in 1979, with Mori’s spectacular proof of a conjecture of Hartshorne characterizing projective spaces as the only smooth projective varieties with ample tangent bundle. The techniques that Mori introduced to solve this conjecture have turned out to have more far-reaching applications than Hartshorne’s conjecture itself.

Recall that a 1-cycle is a finite formal sum $\sum_i n_i [C_i]$ of integral curves, with $n_i \in \mathbb{Z}$. It is rational if its components are rational curves. For a proper morphism, pushforward includes both multiplicity and mapping degree:

$$f_* \left(\sum_i m_i [C_i] \right) = \sum_{\dim f(C_i)=1} m_i \deg(C_i/f(C_i)) [f(C_i)].$$

Contracted components contribute zero. In particular, a nonreduced fiber must be taken with its scheme-theoretic multiplicities.

The original bend-and-break lemma says that a curve moving nontrivially as a morphism while keeping one point image fixed degenerates to an effective cycle with a rational component through that point. In these proofs we resolve rational maps from smooth surfaces by point blow-ups; this supplies rational exceptional curves directly, without requiring the higher-dimensional Abhyankar lemma.

Lemma 2.2.1 (Bend and Break I). *Let X be projective, C a smooth projective curve, $c_0 \in C$, and $f : C \rightarrow X$ a nonconstant morphism. Assume there are a smooth connected nonproper pointed curve $t_0 \in T$ and a morphism $F : C \times T \rightarrow X$ such that*

- (1) $F|_{C \times \{t_0\}} = f$;
- (2) $F(\{c_0\} \times T) = \{f(c_0)\}$;
- (3) $F|_{C \times \{t\}} \neq f$ for general $t \in T$.

Then there are a possibly constant morphism $f' : C \rightarrow X$ and a nonzero effective rational 1-cycle $Z = \sum_i a_i [Z_i]$ such that

$$f_* C \sim_{\text{alg}} f'_* C + Z, \quad f(c_0) \in \text{Supp } Z.$$

In particular, X contains a rational curve through $f(c_0)$.

Remark 2.2.2. If X is smooth along $f(C)$, Section 2.1E shows that it suffices to have

$$-(K_X \cdot f_* C) - g(C) \dim X \geq 1.$$

Indeed a positive-dimensional component through $[f]$ contains a nonconstant curve through that point; normalization and restriction to an open subset give the required smooth parameter curve.

Proof of Lemma 2.2.1. Let $\bar{T}$ be the smooth compactification and $F' : C \times \bar{T} \dashrightarrow X$ the induced rational map. We claim it is undefined at some point of $\{c_0\} \times \bar{T}$.

Suppose it were defined along that whole curve. Its indeterminacy locus is closed, and projection to C is proper, so there is an open neighborhood U of c_0 such that F' is a morphism on $U \times \bar{T}$. The fiber $\{c_0\} \times \bar{T}$ is contracted. By the rigidity lemma, after shrinking U if necessary, F' is constant on each fiber of $U \times \bar{T} \rightarrow U$. Hence

$$F'(c, t) = F'(c, t_0) = f(c) \quad (c \in U).$$

For each $t \in T$, the two morphisms $C \rightarrow X$ agree on the dense open U , hence everywhere because X is separated. This contradicts (3).

Choose an indeterminacy point $a = (c_0, t)$ and resolve by point blow-ups:

$$\begin{array}{ccc} & S & \\ \varepsilon \swarrow & & \searrow F_S \\ C \times \bar{T} & \dashrightarrow & X \\ & F' & \end{array}$$

Take the resolution to be an isomorphism above the regular locus of F' . The total transform of the special fiber is

$$\varepsilon^*(C \times \{t\}) = C' + \sum_i m_i E_i,$$

where $C' \cong C$ is its strict transform, the E_i are rational exceptional curves, and $m_i > 0$. Put $f' = F_S|_{C'}$ and

$$Z = (F_S)_* \left(\sum_i m_i E_i \right).$$

The coefficients in Z include the degrees of the nonconstant maps $E_i \rightarrow F_S(E_i)$. By Lüroth's theorem, each nonconstant image is rational.

The morphism $p : S \rightarrow \bar{T}$ is flat: over a smooth curve, the integral surface has torsion-free, hence flat, local modules. Its fibers, with multiplicities, are algebraically equivalent. Pushing forward and comparing with t_0 gives

$$f_* C \sim_{\text{alg}} f'_* C + Z.$$

It remains to prove both $Z \neq 0$ and the marked-point assertion. The connected exceptional tree $\varepsilon^{-1}(a)$ cannot be entirely contracted by F_S . If it were, the graph of F' would have a single point over a ; its projection to the normal surface $C \times \bar{T}$ would be finite and birational near a , hence an isomorphism. This would extend F' across a , a contradiction.

The strict transform of $\{c_0\} \times \bar{T}$ meets this tree and maps constantly to $f(c_0)$. Follow a path in the tree from that intersection to the first component with nonconstant image. All preceding contracted components map to the same point $f(c_0)$. Thus this first nonconstant image contains $f(c_0)$ and occurs in Z with positive coefficient. This proves the two assertions.

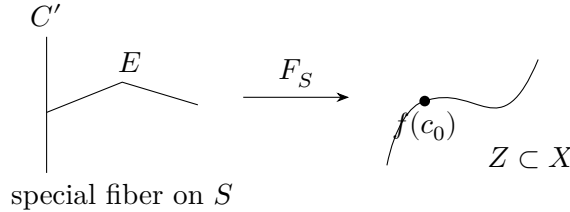


Figure 2.7: Bend and Break I: a nonconstant exceptional component supplies a rational curve through the fixed point.

□

Remark 2.2.3 (Thanks to YuHong Zhou). Under (2) and (3), the curve T cannot be proper. If it were, rigidity applied to $C \times T \rightarrow C$ would give

$$F(c, t) = F(c, t_0) = f(c),$$

contradicting (3). Thus the phrase “possibly nonproper” in [KM98] is harmless: the other hypotheses already exclude the proper case. Writing “nonproper” makes this consequence explicit; it does not contradict their statement.

Lemma 2.2.4 (Bend and Break II). *Let X be projective and $f : \mathbb{P}^1 \rightarrow X$ a nonconstant morphism. Assume there are a smooth connected pointed curve $t_0 \in T$ and a morphism $F : \mathbb{P}^1 \times T \rightarrow X$ such that*

- (1) $F|_{\mathbb{P}^1 \times \{t_0\}} = f$;
- (2) $F(\{0\} \times T) = \{f(0)\}$ and $F(\{\infty\} \times T) = \{f(\infty)\}$;
- (3) *the image of F has dimension two.*

Then $f_\mathbb{P}^1$ is algebraically equivalent to a nonintegral effective rational 1-cycle Z whose support contains $f(0)$ and $f(\infty)$. Thus Z is reducible, or is a multiple rational curve $a[\Gamma]$ with $a > 1$.*

Remark 2.2.5. The condition

$$\dim_{[f]} \text{Mor}(\mathbb{P}^1, X; f|_{\{0, \infty\}}) \geq 2$$

is sufficient for the *conclusion* of the lemma. To see this, let $\Gamma = f(\mathbb{P}^1)_{\text{red}}$ and $d = [k(\mathbb{P}^1) : k(\Gamma)]$. The nonconstant morphism to its image is finite. If $d > 1$, then $f_*\mathbb{P}^1 = d[\Gamma]$ already has the required form.

If $d = 1$, f is the normalization of its image. Morphisms with this same image and the same degree differ by an automorphism of its normalization. Fixing the two point images leaves only finitely many choices for their preimages on the normalization, and for each such choice the remaining automorphism group is one-dimensional, isomorphic to $\mathbb{G}_m$. Hence the fixed-image locus through $[f]$ has dimension at most one. A component of dimension at least two therefore contains a parameter curve whose images actually move, so condition (3) holds for that family.

For multiple covers, variations with the same image need not come from $\mathbb{G}_m$; hence the dimension condition alone does not imply (3) for an arbitrary f . The degree d here is the function-field degree, including any inseparable part; connected fibers are not an additional case outside this definition.

If X is smooth along $f(\mathbb{P}^1)$, Section 2.1E shows that

$$-(K_X \cdot f_*\mathbb{P}^1) - \dim X \geq 2$$

is sufficient for the same conclusion.

Proof of Lemma 2.2.4. Compactify T to $\bar{T}$. Take a $\mathbb{P}^1$ -bundle $\pi : P \rightarrow \bar{T}$ containing $\mathbb{P}^1 \times T$ with its given projection, and resolve the induced rational map $F' : P \dashrightarrow X$ by a sequence of point blow-ups

$$S = S_r \rightarrow S_{r-1} \rightarrow \cdots \rightarrow S_1 \rightarrow P, \quad F_S : S \rightarrow X.$$

Among all such bundle models and resolutions, choose one with the number r minimal. We may take each resolution to be an isomorphism over the regular locus of the rational map. Put $p = \pi \circ \varepsilon : S \rightarrow \bar{T}$.

Each fiber $p^{-1}(t)$ is a connected tree of rational curves, with its total-transform multiplicities. As in the preceding lemma, the cycles

$$Z_t = (F_S)_*[p^{-1}(t)]$$

are effective, rational, and algebraically equivalent to $f_*\mathbb{P}^1$. The closures of the two marked sections lift to sections of p and map to $f(0)$ and $f(\infty)$. Thus these points belong to the image of each fiber. They also belong to $\text{Supp } Z_t$: the fiber has some nonconstant component because its ample degree equals the positive degree of $f_*\mathbb{P}^1$, and a connected chain of contracted components attaches to such a component with the same image point.

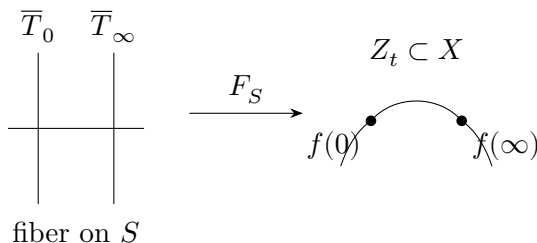


Figure 2.8: The support of the fiber cycle contains both marked images.

Assume, seeking a contradiction, that every Z_t is integral. There is then a unique component R of each fiber that is not contracted by F_S , and

$$1 = \text{mult}_R(p^{-1}(t)) \deg(R/F_S(R)).$$

Both positive integers are therefore one.

The case $r = 0$

Now $F' : P \rightarrow X$ is a morphism. For an ample divisor H on X , put $L = F'^*H$. The image has dimension two, so $L^2 > 0$. Both distinct sections $\bar{T}_0, \bar{T}_\infty$ are contracted, hence

$$L \cdot \bar{T}_0 = L \cdot \bar{T}_\infty = 0.$$

Hodge index gives $\bar{T}_0^2 < 0$ and $\bar{T}_\infty^2 < 0$. Their numerical classes are independent: proportionality, intersected with a fiber, would force them to be equal, contrary to their negative self-intersection and $\bar{T}_0 \cdot \bar{T}_\infty \geq 0$. Together with L they would give three independent classes in $N^1(P)_\mathbb{R}$. This contradicts $\rho(P) = 2$ for a ruled surface [Har77, V, 2.3].

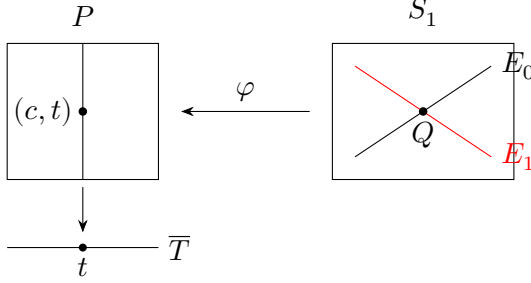
The case $r > 0$

Consider the first blow-up $\varphi : S_1 \rightarrow P$, at a point on the fiber over t . Its fiber has two components E_0, E_1 meeting transversally at Q , with E_0 the strict

transform of the original fiber and E_1 exceptional. Both are (-1) -curves, since

$$E_0^2 = (\varphi^*[\pi^{-1}(t)] - E_1)^2 = 0 - 1 = -1.$$

Write $\varepsilon_1 : S \rightarrow S_1$ for the remaining blow-ups.



By minimality, some component of $\varepsilon_1^{-1}(E_1)$ is not contracted by F_S . Otherwise this connected inverse image would map to a point, and descent through the normal surface would extend F' over the first center, making that blow-up unnecessary. Thus the unique noncontracted component R lies in $\varepsilon_1^{-1}(E_1)$.

Every exceptional component above the crossing Q has fiber multiplicity at least two. The first blow-up of Q has multiplicity $1 + 1 = 2$; a later blow-up either preserves a component's multiplicity at a smooth point or adds multiplicities at an intersection. Therefore R , whose multiplicity is one, cannot lie over Q . It follows that all of $\varepsilon_1^{-1}(E_0)$ is contracted by F_S , and, by connectedness, to a single point.

The same graph-descent argument shows that the induced rational map $S_1 \dashrightarrow X$ is regular along E_0 . A minimal resolution has no further blow-up centers there. We may now contract E_0 on S_1 . This is the elementary transformation of a ruled surface: it gives another $\mathbb{P}^1$ -bundle $P' \rightarrow \bar{T}$, identical to P over T . Since E_0 is contracted by the map to X , that map descends near its contraction point. The remaining blow-ups, disjoint from E_0 , give a resolution over P' using $r - 1$ blow-ups. This contradicts the choice of r .

Thus some Z_t is nonintegral, proving the lemma. $\square$

Remark 2.2.6. The two fixed domain points can be any distinct $q = [q_1 : q_2]$ and $q' = [q'_1 : q'_2]$. The matrix

$$\begin{pmatrix} q'_1 & q_1 \\ q'_2 & q_2 \end{pmatrix} \in \mathrm{PGL}_2(k)$$

sends $0 = [0 : 1]$ to q and $\infty = [1 : 0]$ to q' . Reparametrization gives the same conclusion: the support of the resulting cycle contains $f(q)$ and $f(q')$. These points need not lie on different components, and the argument does not require simultaneous indeterminacy on both marked sections.

If all ε -exceptional components of a particular fiber are contracted, the push-forward of that fiber comes from its strict transform $R_0 \cong \mathbb{P}^1$, which has fiber multiplicity one. If its cycle is nonintegral, then

$$(F_S)_*[p^{-1}(t)] = d[\Gamma], \quad d = \deg(F_S|_{R_0} : R_0 \rightarrow \Gamma) > 1.$$

This is a sufficient mechanism for a multiple curve, but not the only one: several noncontracted components may have the same image and their coefficients add. The inequality concerns the map on this special component; it does not imply that the original f has degree greater than one onto its own image.

For a stronger bend-and-break statement with more fixed points, see [Deb01, 3.5].

2.3 Application: Finding Rational Curves

2.3A Rational Curves on Fano Varieties

We are ready to formulate and prove Mori's result on rational curves on Fano varieties. Here a Fano variety is smooth, projective, of positive dimension, with ample anticanonical divisor.

For this section and Section 2.4 the original ground field k is algebraically closed of characteristic zero. The symbols X, C denote the original varieties over k . Models over finitely generated rings will be written $\mathcal{X}, \mathcal{C}$, and their geometric reductions in positive characteristic will be denoted X_p, C_p or X_P, C_P . This convention is maintained when applying Frobenius.

Example 2.3.1 (The simplest Fano varieties). For a smooth projective hypersurface $X_d \subset \mathbb{P}^{n+1}$ of degree d , with $n \geq 1$, adjunction gives

$$K_{X_d} = (K_{\mathbb{P}^{n+1}} + dH)|_{X_d} = (d - n - 2)H|_{X_d}.$$

Thus X_d is Fano exactly when $d < n + 2$.

Example 2.3.2 (Constructing Fano varieties). Let Y be Fano and $D_1, \dots, D_r$ nef divisors such that

$$M = -K_Y - D_1 - \dots - D_r$$

is ample. Then

$$X = \mathbb{P}_Y \left(\bigoplus_{i=1}^r \mathcal{O}_Y(D_i) \right)$$

is Fano. We use the quotient convention for the projective bundle and write D for the class of $\mathcal{O}_X(1)$.

For $\pi : X \rightarrow Y$, the relative Euler sequence [Har77, III, Exercise 8.4(b)] is

$$0 \rightarrow \Omega_{X/Y} \rightarrow \pi^* \left(\bigoplus_i \mathcal{O}_Y(D_i) \right) \otimes \mathcal{O}_X(-1) \rightarrow \mathcal{O}_X \rightarrow 0.$$

Taking determinants gives

$$\det \Omega_{X/Y} \cong \pi^* \mathcal{O}_Y \left(\sum_i D_i \right) \otimes \mathcal{O}_X(-r).$$

The map π is smooth because locally it is the projection of a projective-space bundle; on its standard affine charts it is a polynomial extension. Thus

$$0 \rightarrow \pi^* \Omega_Y \rightarrow \Omega_X \rightarrow \Omega_{X/Y} \rightarrow 0$$

is exact, and

$$-K_X \sim_{\text{lin}} rD + \pi^* M.$$

Each $M + D_i$ is ample. The vector bundle $\bigoplus_i \mathcal{O}_Y(M + D_i)$ is ample [Laz04b, 6.1.13]. Tensoring a bundle by a line bundle leaves its projectivization unchanged, with

$$\begin{aligned} \varphi : X' = \mathbb{P}_Y \left(\bigoplus_i \mathcal{O}_Y(M + D_i) \right) &\xrightarrow{\sim} X, \\ \mathcal{O}_{X'}(1) &\cong \varphi^*(\mathcal{O}_X(1) \otimes \pi^* \mathcal{O}_Y(M)). \end{aligned}$$

Hence $D + \pi^* M$ is ample. Moreover D is nef, since the tautological quotient of the pullback of a direct sum of nef line bundles has nonnegative degree on every curve. More explicitly, on the normalization of a curve in X , some summand maps nontrivially to that quotient line bundle, so its degree is at least the degree of that summand, which is nonnegative. Therefore

$$rD + \pi^* M = (D + \pi^* M) + (r-1)D$$

is ample, and X is Fano.

Lemma 2.3.3. *Let R be a finitely generated $\mathbb{Z}$ -algebra.*

- (1) *For every maximal ideal $\mathfrak{m} \subset R$, the residue field $R/\mathfrak{m}$ is finite.*
- (2) *Closed points are Zariski dense in $\text{Spec } R$.*

Proof. (1) Put $L = R/\mathfrak{m}$. If L had characteristic zero, it would be a finitely generated $\mathbb{Q}$ -algebra and a field, hence a finite extension of $\mathbb{Q}$ by Zariski's lemma. Let $x_1, \dots, x_s$ generate it as a $\mathbb{Z}$ -algebra. Choose $N > 0$ clearing the denominators in monic algebraic equations for these elements over $\mathbb{Q}$. Then each x_i is integral over $\mathbb{Z}[1/N]$, so $L = \mathbb{Z}[1/N][x_1, \dots, x_s]$ is integral over $\mathbb{Z}[1/N]$. A subring over which a field is integral is itself a field, but $\mathbb{Z}[1/N]$ is not a field. This is a contradiction.

Thus $\text{char } L = p > 0$. The field L is a finitely generated algebra over $\mathbb{F}_p$, hence is a finite extension by Zariski's lemma, so it is finite.

(2) Take a nonempty basic open $D(a) = \text{Spec } R_a$. Nonemptiness, rather than merely $a \neq 0$, is needed if R has nilpotents. Choose a maximal ideal $\mathfrak{n}$ of R_a . Its

residue field is finite by (1). If $\mathfrak{m}$ is the inverse image in R , then $R/\mathfrak{m}$ embeds in that finite field and is a finite domain, hence a field. Thus $\mathfrak{m}$ is a closed point in $D(a)$. The same argument applied to any quotient of R shows that closed points are dense in every closed subscheme as well. $\square$

Lemma 2.3.4. *A homogeneous system of equations with integer coefficients which has a nontrivial solution over $\overline{\mathbb{F}}_p$ for infinitely many primes p has a nontrivial solution over every algebraically closed field of characteristic zero.*

More generally, let R be a finitely generated integral $\mathbb{Z}$ -algebra of characteristic zero, and take homogeneous equations with coefficients in R . If their projective zero scheme has nonempty geometric fibers over a Zariski dense subset of $\text{Spec } R$, then it has a point over every algebraically closed field containing $\text{Frac } R$, with the given embedding of R .

Proof. The equations define a closed subscheme $Z \subset \mathbb{P}_R^N$. Since $Z \rightarrow \text{Spec } R$ is proper, its image is closed. It contains the given dense subset, so it is all of $\text{Spec } R$. In particular, the generic fiber Z_η is nonempty.

Let $K = \text{Frac } R$. A closed point of the finite type K -scheme Z_η has residue field finite over K . This field embeds over K into any algebraically closed extension of K , giving the required point after base change. For the original integer-coefficient statement, take $R = \mathbb{Z}$: infinitely many closed points are dense, and $K = \mathbb{Q}$ embeds into every characteristic-zero field. For a general R , the condition that the chosen algebraically closed field contain $\text{Frac } R$ cannot be omitted; for example, $\mathbb{Q}(t)$ cannot embed into $\overline{\mathbb{Q}}$. $\square$

Remark 2.3.5 (The functor-of-points view). For an R -algebra field K ,

$$\mathbb{P}_R^N(K) = \text{Hom}_R(\text{Spec } K, \mathbb{P}_R^N) \cong (K^{N+1} \setminus \{0\})/K^*.$$

Thus a projective point is a nonzero tuple up to simultaneous nonzero scaling. On an affine chart, the corresponding description is

$$\text{Hom}_{R\text{-alg}}(R[T_1, \dots, T_N]/(f_1, \dots, f_r), A) \cong \{(a_1, \dots, a_N) \in A^N \mid f_j(a) = 0 \text{ for all } j\}.$$

For $Z = V_+(F_1, \dots, F_r)$ this gives

$$Z(K) = \{[x_0 : \dots : x_N] \in \mathbb{P}^N(K) \mid F_j(x) = 0 \text{ for all } j\}.$$

If K is algebraically closed, nonemptiness of the finite type fiber Z_K is equivalent to the existence of a K -point, by the Nullstellensatz. The algebraic-closedness condition matters.

The same strategy also applies to open conditions such as the absence of a common zero in the coordinates of a morphism. For those parameter spaces, properness need not hold. Instead one uses Chevalley's theorem: the image of a finite type morphism is constructible. A dense constructible subset of an integral noetherian base contains a nonempty open subset, hence its generic point. This is the form used in Step 8 of Theorem 2.3.6.

Theorem 2.3.6 ([Mor82]). *For every point x of a Fano variety X , there is a rational curve D through x such that*

$$0 < -(D \cdot K_X) \leq \dim X + 1.$$

Proof. This is done in several steps. Write $n = \dim X$.

Step 1: To find this rational curve, we intend to use the bend-and-break technique. Thus we have to find a morphism $f : C \rightarrow X$ which we will be able to deform, then we bend these deformations together over a fixed point, and break them into some rational curves.

Pick an integral projective curve $C' \subseteq X$ passing through x , and let $f : C \rightarrow C' \hookrightarrow X$ be its normalization. Then C is a smooth projective curve. Pick $c_0 \in C$ such that $f(c_0) = x$, which is the target point we are bending.

Step 2: According to the formula in Section 2.1E,

$$\dim_{[f]} \text{Mor}(C, X; f|_{\{c_0\}}) \geq -(K_X \cdot f_*C) - g(C) \cdot n.$$

Thus whenever the quantity $-(K_X \cdot f_*C) - g(C)n$ is positive, there must be a non-trivial one-dimensional family of deformations of the map $f : C \rightarrow X$ keeping the image of c_0 fixed. So using Bend and Break I, we obtain a rational curve in X through x .

Step 3: Now we show how to get the positive quantity

$$-(K_X \cdot f_*C) - g(C) \cdot n > 0.$$

To do this, we need to get $-(K_X \cdot f_*C)$ to be big enough. We consider three cases:

(1) $g(C) = 0$.

Then automatically $-(K_X \cdot f_*C) > 0$, since $-K_X$ is ample. Actually, by Lüroth's theorem, the image of C is already a rational curve through x .

(2) $g(C) = 1$.

For the same reason we have $-(K_X \cdot f_*C) > 0$, but as we want it to be big enough, we apply some finite endomorphism of C fixing the point c_0 .

In particular, take c_0 as the origin of the elliptic curve C . Via the injective ring homomorphism [Har77, IV, 4.10]

$$\mathbb{Z} \rightarrow \text{End}(C, c_0),$$

we can pick the multiplication-by- r endomorphism $[r]_C$ and compose it with f . Then

$$-(K_X \cdot (f \circ [r]_C)_*C) - n = -r^2(K_X \cdot f_*C) - n,$$

where the equality follows from $\deg[r]_C = r^2$ [Har77, IV, 4.17]. This is positive for $r \gg 0$, and $(f \circ [r]_C)(c_0) = x$. Thus a multiple of the original cycle admits deformations with the image of one point fixed.

$$(C, c_0) \xrightarrow[\deg=r^2]{[r]_C} (C, c_0) \xrightarrow{f} (X, x).$$

Figure 2.9. The endomorphism $[r]_C$ raises the mapping degree while preserving the genus and the marked point.

(3) $g(C) \geq 2$.

By the Hurwitz formula for a separable endomorphism of degree d , we have

$$2g(C) - 2 = d \cdot (2g(C) - 2) + \deg R_{\text{ram}}.$$

If $d > 1$, the right hand side is strictly larger than the left, since $\deg R_{\text{ram}} \geq 0$. So a curve of genus at least 2 has no separable endomorphism of degree greater than 1. In characteristic zero, every finite endomorphism is separable.

Thus we can try to consider another smooth projective curve $h : C'' \rightarrow C$ and deform $f \circ h$. But the problem here is that even though

$$((f \circ h)_* C'' \cdot K_X) = \deg h \cdot (f_* C \cdot K_X),$$

the genus also changes. For example, if C'' is a connected r -sheeted unbranched cover of C , then by the Hurwitz formula again,

$$2g(C'') - 2 = r \cdot (2g(C) - 2).$$

Hence

$$\begin{aligned} & - (K_X \cdot (f \circ h)_* C'') - g(C'') \cdot n \\ & = -r(K_X \cdot f_* C) + g(C) \cdot n + (r - 1)n. \end{aligned}$$

This does not necessarily become positive even by making r large enough.

Step 4: Thus we are only in trouble in the case $g(C) > 1$, because in characteristic zero C does not admit an endomorphism of high degree. To overcome this, we will first reduce the data to positive characteristic. In the situation of Step 5, write C_p for the smooth projective reduction of C over $\overline{\mathbb{F}}_p$, defined over $\mathbb{F}_p$. Raising its homogeneous coordinates to the p -th power gives an $\overline{\mathbb{F}}_p$ -endomorphism

$$F_p : C_p \rightarrow C_p, \quad \deg F_p = p.$$

Its iterates $F_p^m : C_p \rightarrow C_p$ have degree p^m , while the domain remains the same curve C_p and hence has the same genus. Thus Frobenius will be applied to the reduction C_p , not to the original characteristic-zero curve C .

Next we construct these reductions. Step 6 treats the general case, where the residue field may be $\mathbb{F}_q$ and we use the q -power Frobenius on C_P .

Step 5: First suppose that X , C , the morphism f , and the point c_0 are all defined over $\mathbb{Q}$. After clearing denominators, take equations with integral coefficients

$$\begin{aligned} h_1(x_0, \dots, x_N), \dots, h_a(x_0, \dots, x_N) & \text{ for } X, \\ c_1(y_0, \dots, y_M), \dots, c_b(y_0, \dots, y_M) & \text{ for } C. \end{aligned}$$

We also spread out the graph of f and the marked point c_0 . After inverting finitely many primes, these data give models

$$\mathcal{X} \longrightarrow U, \quad \mathcal{C} \longrightarrow U, \quad f_U : \mathcal{C} \longrightarrow \mathcal{X}, \quad U = \operatorname{Spec} \mathbb{Z}[1/N_0],$$

together with a section $c_{0,U} : U \longrightarrow \mathcal{C}$, meaning that its composite with $\mathcal{C} \rightarrow U$ is id_U . It selects a marked point on each fibre. Set $x_U = f_U \circ c_{0,U}$. We may shrink U so that $\mathcal{X} \rightarrow U$ and $\mathcal{C} \rightarrow U$ are smooth and projective with geometrically integral fibres, of relative dimensions n and 1 respectively.

The sheaves

$$\mathcal{O}_{\mathcal{C}}, \quad f_U^* \omega_{\mathcal{X}/U}, \quad f_U^* T_{\mathcal{X}/U}$$

are vector bundles on $\mathcal{C}$ and are flat over U . Since $\mathcal{C} \rightarrow U$ is proper, their fibrewise Euler characteristics are locally constant. For instance, writing $f_p : C_p \rightarrow X_p$ for the geometric fibre over (p) ,

$$(K_{X_p} \cdot (f_p)_* C_p) = \deg f_p^* \omega_{X_p} = \chi(f_p^* \omega_{X_p}) - \chi(\mathcal{O}_{C_p}).$$

As U is connected, the quantities

$$(K_{X_p} \cdot (f_p)_* C_p), \quad g(C_p), \quad \chi(f_p^* T_{X_p})$$

are constant, and agree with their values on the original curve.

Moreover, ω_X^{-1} extends to $\omega_{X/U}^{-1}$. By openness of ampleness on the fibres of a proper morphism, we may shrink U again so that $-K_{X_p}$ is ample for every geometric fibre. Hence

$$A := -(K_{X_p} \cdot (f_p)_* C_p) = -(K_X \cdot f_* C) > 0.$$

The geometric fibres C_p and X_p are defined over $\mathbb{F}_p$. Raising the homogeneous coordinates of C_p to the p -th power gives the Frobenius endomorphism F_p , a finite morphism of degree p . Since the section $c_{0,U}$ was included in the data, $c_{0,p}$ is $\mathbb{F}_p$ -rational and $F_p^m(c_{0,p}) = c_{0,p}$. Therefore the base-pointed deformation space of the composite morphism

$$C_p \xrightarrow{F_p^m} C_p \xrightarrow{f_p} X_p$$

has dimension bounded below by

$$\begin{aligned} \dim_{[f_p \circ F_p^m]} \operatorname{Mor}(C_p, X_p; (f_p \circ F_p^m)|_{\{c_{0,p}\}}) \\ \geq -p^m (K_{X_p} \cdot (f_p)_* C_p) - g(C_p) \cdot n \\ = p^m A - g(C)n. \end{aligned}$$

This is positive for $m \gg 0$. Then as in Step 2, we produce a rational curve A_p on X_p through x_p for all primes represented in U .

Step 6: Suppose now that we are in the general case. The coefficients of the h_j defining X , the c_j defining C , the equations defining the graph $\Gamma(f)$, and the coordinates of c_0 need not be rational. These finitely many coefficients lie in a finitely generated $\mathbb{Z}$ -subalgebra $R \subseteq k$. After adjoining the necessary coefficients and localizing R , we obtain models and sections as in Step 5 over a non-empty affine open subset $U \subseteq \text{Spec } R$. We may again arrange smoothness, projectivity, geometric integrality, constancy of the above numerical quantities, and ampleness of the anticanonical bundles on the geometric fibres.

Now let $P \in U$ be a closed point. By Lemma 2.3.3, its residue field $\kappa(P)$ is finite. Set $q = |\kappa(P)|$, and denote the geometric fibres by C_P and X_P . The q -power Frobenius endomorphism $F_q : C_P \rightarrow C_P$ fixes $c_{0,P}$, and its m -th iterate has degree q^m . Thus

$$\begin{aligned} \dim_{[f_P \circ F_q^m]} \text{Mor}(C_P, X_P; (f_P \circ F_q^m)|_{\{c_{0,P}\}}) \\ \geq -q^m(K_{X_P} \cdot (f_P)_* C_P) - g(C_P) \cdot n > 0 \end{aligned}$$

for $m \gg 0$. This gives a rational curve A_P through x_P for every closed point $P \in U$.

Step 7: In this step, we wish to obtain the required inequality, so we wish to replace A_P with another rational curve B_P through x_P such that

$$0 < -(K_{X_P} \cdot B_P) \leq n + 1.$$

To do this, notice that if $-(K_{X_P} \cdot A_P) > n + 1$, then we can take the normalization morphism

$$\varphi : \mathbb{P}^1_{\kappa(P)} \rightarrow A_P \hookrightarrow X_P.$$

In particular, $\varphi_* \mathbb{P}^1 = A_P$ as 1-cycles. Choose $q_0 \in \mathbb{P}^1$ with $\varphi(q_0) = x_P$, and choose $q_1 \neq q_0$. According to Section 2.1E, with these two point images fixed, we have

$$\begin{aligned} \dim_{[\varphi]} \text{Mor}(\mathbb{P}^1, X_P; \varphi|_{\{q_0, q_1\}}) &\geq -(K_{X_P} \cdot \varphi_* \mathbb{P}^1) - n \\ &= -(K_{X_P} \cdot A_P) - n > 1. \end{aligned}$$

So this deformation space has dimension at least two.

But notice that φ is birational onto its image. Parametrizations with the same image differ by an automorphism of its normalization. Keeping the two point images fixed allows only finitely many choices of their preimages on the normalization; for each such choice, the remaining automorphisms form a one-dimensional family. Hence the fixed-image locus near $[\varphi]$ has dimension at most one, and we can choose a one-dimensional family in which the image of $\mathbb{P}^1$ actually moves.

By Bend and Break II, we obtain

$$A_P = \varphi_* \mathbb{P}^1 \sim_{\text{alg}} \sum_i a_i C_i, \quad a_i \in \mathbb{Z}_{>0}, \quad x_P \in \bigcup_i C_i,$$

where the C_i are integral rational curves and the effective cycle on the right is non-integral: it is reducible or a multiple rational curve. Since $-K_{X_P}$ is ample, each component satisfies

$$0 < -(K_{X_P} \cdot C_i) < -(K_{X_P} \cdot A_P).$$

This strict inequality also holds when the right hand side is a single multiple curve aC with $a > 1$.

Choose a component passing through x_P , take its normalization, and continue this process of deformation and decomposition. As long as the anticanonical degree is greater than $n + 1$, we find a rational curve through x_P of strictly smaller positive integral degree. This process must terminate in finitely many steps, giving our desired rational curve B_P with

$$x_P \in B_P, \quad 0 < -(K_{X_P} \cdot B_P) \leq n + 1.$$

Step 8: In this step, we must conclude the existence of a rational curve on the original Fano variety X , in particular in characteristic zero, from the existence of bounded degree rational curves B_P for every closed point $P \in U$.

Choose a positive integer ℓ such that $-\ell K_X$ is very ample, and fix the corresponding embedding $X \hookrightarrow \mathbb{P}_k^N$. By enlarging R and shrinking U if necessary, spread out this fixed embedding to

$$\mathcal{X} \hookrightarrow \mathbb{P}_U^N, \quad \mathcal{O}_{\mathcal{X}}(1) \simeq \omega_{\mathcal{X}/U}^{-\ell}.$$

All the preceding arguments apply to these models. Since a closed immersion remains a closed immersion after base change, this gives embeddings of all the geometric fibres into a fixed projective space. The degree of B_P under this embedding is

$$-\ell(K_{X_P} \cdot B_P) \leq \ell(n + 1).$$

Set $D_0 = \ell(n + 1)$. For $1 \leq e \leq D_0$, consider the relative space of morphisms of degree e , measured by this embedding, with the image of $0 = [0 : 1] \in \mathbb{P}^1$ fixed at x_U . Their disjoint union is

$$\mathcal{M} = \coprod_{e=1}^{D_0} \text{Mor}_e(\mathbb{P}_U^1, \mathcal{X}; 0 \mapsto x_U).$$

This is a scheme of finite type over U . On each geometric fibre, Mor_e parametrizes tuples of homogeneous polynomials of degree e , up to simultaneous non-zero scaling, with no common zero, satisfying the equations of the target and the prescribed

point condition. Thus these are actual morphisms, and $e \geq 1$ excludes constant maps. We allow all degrees between 1 and D_0 instead of multiplying coordinates by a common factor.

For every closed point $P \in U$, the normalization of B_P , after reparametrizing so that 0 maps to x_P , gives a geometric point of $\mathcal{M}_P$. By Chevalley's theorem, the image of $\mathcal{M} \rightarrow U$ is constructible. Since U is an integral scheme of finite type over $\mathbb{Z}$, its closed points are dense. This constructible image contains every closed point, so it is dense. A dense constructible subset of an integral noetherian scheme contains a non-empty open subset, hence contains the generic point η of U .

Therefore $\mathcal{M}_\eta$ is non-empty. Write $K = \text{Frac } R \subseteq k$. A closed point of $\mathcal{M}_\eta$ has residue field finite over K , and this field embeds into the original algebraically closed field k over K . After base change, we obtain a non-constant morphism

$$g : \mathbb{P}_k^1 \rightarrow X, \quad g(0) = x, \quad 1 \leq \deg g^* \mathcal{O}_X(1) \leq D_0.$$

Here we return to the original field k containing R ; no embedding of $\text{Frac } R$ into an arbitrary algebraically closed field is required.

Step 9: Let $D = g(\mathbb{P}^1)_{\text{red}}$. By Lüroth's theorem, D is a rational curve, and $x \in D$. If δ is the degree of g onto its image, then $g_* \mathbb{P}^1 = \delta D$, so

$$\ell \delta (-(K_X \cdot D)) = \deg g^* \mathcal{O}_X(1) \leq \ell(n+1).$$

Hence $-(K_X \cdot D) \leq n+1$. This number is strictly positive since $-K_X$ is ample. Thus D satisfies the required inequality. $\square$

2.3B Finding Rational Curves when K_X is not Nef

Now let us weaken our hypothesis on the smooth projective variety X . We no longer require $-K_X$ to be ample; we only assume that K_X is not nef, which means that there is an integral curve $C' \subseteq X$ such that

$$-(K_X \cdot C') > 0.$$

This imposes no nefness assumption on $-K_X$. We also fix an ample Cartier divisor H on X .

Theorem 2.3.7 ([Mor82]). *Let X be a smooth projective variety and H an ample Cartier divisor on X . Assume that there is an integral curve $C' \subseteq X$ such that $-(C' \cdot K_X) > 0$. Then there is a rational curve $E \subseteq X$ such that*

$$0 < -(E \cdot K_X) \leq \dim X + 1, \quad \frac{-(E \cdot K_X)}{E \cdot H} \geq \frac{-(C' \cdot K_X)}{C' \cdot H}.$$

Proof. For finding rational curves, we can use a similar technique as before, passing to fields of positive characteristic to ensure that the dimension of the deformation space is large enough. Write $n = \dim X$.

Step 1: Let $f : C \rightarrow C' \hookrightarrow X$ be the normalization of C' . It is birational onto its image and $f_*C = C'$ as 1-cycles. Choose $c_0 \in C$. If

$$\dim_{[f]} \text{Mor}(C, X; f|_{\{c_0\}}) \geq -(K_X \cdot f_*C) - g(C) \cdot n > 0,$$

then f can deform with the image of c_0 fixed. As before, using Bend and Break I, we obtain

$$f_*C \sim_{\text{alg}} f'_*C + Z,$$

where $f' : C \rightarrow X$ may be constant and Z is a non-zero effective rational 1-cycle. We keep the pushforward cycles here, since the remaining morphism need not be birational onto its image.

Step 2 (Repeating bend and break): In order to achieve the above lower bound on the deformation space, we pass to positive characteristic as in Steps 5–6 of Theorem 2.3.6. Spread out X, C, f, c_0 , and also H , over a non-empty affine integral scheme U of finite type over $\mathbb{Z}$. After shrinking U , take smooth projective models $\mathcal{X}, \mathcal{C}$, a morphism $f_U : \mathcal{C} \rightarrow \mathcal{X}$ extending f , and a relatively ample line bundle $\mathcal{H}$ extending $\mathcal{O}_X(H)$.

For a closed point $P \in U$, use the same notation $f_P : C_P \rightarrow X_P$ for the geometric fibres, and write H_P for the corresponding divisor class. Set

$$A = -(K_{X_P} \cdot (f_P)_*C_P), \quad B = H_P \cdot (f_P)_*C_P, \quad G = g(C_P)n.$$

These quantities are independent of P , with $A, B > 0$. In particular,

$$M := \frac{-(C' \cdot K_X)}{C' \cdot H} = \frac{-(K_{X_P} \cdot (f_P)_*C_P)}{H_P \cdot (f_P)_*C_P} = \frac{A}{B} > 0.$$

Here constancy follows from the same Euler characteristic argument as in Theorem 2.3.6, applied also to $f_U^*\mathcal{H}$.

Put $q = |\kappa(P)|$, and compose f_P with the m -th iterate of the q -power Frobenius $F_q : C_P \rightarrow C_P$. Writing $Q = q^m$, we get

$$\begin{aligned} \dim_{[f_P \circ F_q^m]} \text{Mor}(C_P, X_P; (f_P \circ F_q^m)|_{\{c_{0,P}\}}) \\ \geq -Q(K_{X_P} \cdot (f_P)_*C_P) - G = QA - G, \end{aligned}$$

which is positive for $m \gg 0$. The ratio M does not change after this composition, since both intersection numbers are multiplied by Q .

Set $f_{P,m}^0 = f_P \circ F_q^m$ and

$$Y_j = (f_{P,m}^j)_*C_P, \quad Y_0 = Q(f_P)_*C_P.$$

Whenever $-(K_{X_P} \cdot Y_{j-1}) > G$, we can deform the remaining morphism with the image of $c_{0,P}$ fixed and apply Bend and Break I again, without composing with Frobenius again. This gives

$$Y_{j-1} \sim_{\text{alg}} Y_j + W_j,$$

where W_j is a non-zero effective rational cycle. The image of the marked point is fixed for the deformation at each individual stage; we do not need it to remain the same between stages.

Recall that algebraic equivalence implies numerical equivalence [Ful98, 10.3]. Since H_P is ample, at each step

$$H_P \cdot Y_j = H_P \cdot Y_{j-1} - H_P \cdot W_j < H_P \cdot Y_{j-1}.$$

These are non-negative integers, so the process must stop after finitely many steps, with

$$Q(f_P)_* C_P \sim_{\text{alg}} Y_s + Z_s, \quad Z_s = \sum_{j=1}^s W_j, \quad -(K_{X_P} \cdot Y_s) \leq G.$$

This is our stopping condition; it does not assert that the remaining morphism has no deformations. The cycle Y_s may be zero.

Let

$$\begin{aligned} a &= -(K_{X_P} \cdot Y_s), & b &= -(K_{X_P} \cdot Z_s), \\ c &= H_P \cdot Y_s, & d &= H_P \cdot Z_s. \end{aligned}$$

Then

$$a \leq G, \quad c \geq 0, \quad a + b = QA, \quad c + d = QB.$$

For large m , it follows that $b \geq QA - G > 0$ and $0 < d \leq QB$. We only need the upper bound on a ; no lower bound on a is assumed.

Step 3: We claim the following. For any $\varepsilon > 0$, if $m \gg 0$, then there exists an integral component E_P of the rational cycle Z_s such that

$$\frac{-(E_P \cdot K_{X_P})}{E_P \cdot H_P} > M - \varepsilon.$$

To see this, we use the following lemma.

Lemma 2.3.8. *If $c, d > 0$ and $a, b \in \mathbb{R}$, then*

$$\frac{a + b}{c + d} \leq \max \left\{ \frac{a}{c}, \frac{b}{d} \right\}.$$

Indeed, $(a + b)/(c + d)$ is the weighted average of a/c and b/d , with positive weights c and d . The same argument applies to any finite number of terms.

In our situation, for Q sufficiently large, we have

$$\frac{b}{d} \geq \frac{QA - G}{QB} = M - \frac{G}{QB} > M - \varepsilon.$$

This estimate also applies when $c = 0$; there is no need to form a/c . Write

$$Z_s = \sum_i r_i E_{P,i}, \quad r_i \in \mathbb{Z}_{>0},$$

and set $b_i = -(K_{X_P} \cdot E_{P,i})$ and $d_i = H_P \cdot E_{P,i} > 0$. By the weighted-average argument,

$$\max_i \left\{ \frac{b_i}{d_i} \right\} \geq \frac{\sum_i r_i b_i}{\sum_i r_i d_i} = \frac{b}{d} > M - \varepsilon.$$

Thus the required component E_P exists.

Step 4: From now on, take $0 < \varepsilon < M/2$. Suppose that

$$-(E_P \cdot K_{X_P}) > n + 1$$

for the rational curve E_P obtained above. Take its normalization $\varphi : \mathbb{P}^1 \rightarrow E_P \hookrightarrow X_P$ and fix the images of two distinct points $q_0, q_1 \in \mathbb{P}^1$. Then

$$\dim_{[\varphi]} \text{Mor}(\mathbb{P}^1, X_P; \varphi|_{\{q_0, q_1\}}) \geq -(E_P \cdot K_{X_P}) - n > 1.$$

As in Step 7 of Theorem 2.3.6, the fixed-image locus has dimension at most one. Hence we can move the image curve and apply Bend and Break II to obtain a non-integral effective rational cycle

$$E_P \sim_{\text{alg}} \sum_i a_i E'_{P,i}.$$

Using Lemma 2.3.8 with the multiplicities a_i , at least one component E'_P satisfies

$$\frac{-(E'_P \cdot K_{X_P})}{E'_P \cdot H_P} > M - \varepsilon. \quad (2.1)$$

Since $M - \varepsilon > 0$, this component has positive anticanonical degree. Also,

$$0 < H_P \cdot E'_P < H_P \cdot E_P,$$

because H_P is ample and the cycle is non-integral. This includes the case of a multiple rational curve.

If $-(E'_P \cdot K_{X_P}) > n + 1$, take its normalization and repeat. This process cannot continue infinitely, since at each step the positive integer $H_P \cdot E_P$ decreases. Eventually we arrive at a rational curve, again called E_P , such that

$$0 < -(E_P \cdot K_{X_P}) \leq n + 1, \quad \frac{-(E_P \cdot K_{X_P})}{E_P \cdot H_P} > M - \varepsilon.$$

No nefness of $-K_{X_P}$ is needed here: positivity on the selected component follows from its ratio, and termination follows from H_P .

These inequalities imply

$$1 \leq E_P \cdot H_P < \frac{n+1}{M-\varepsilon} < \frac{2(n+1)}{M}.$$

Therefore, for all $0 < \varepsilon < M/2$ and all closed points P , the possible ratios belong to the same finite set

$$\mathcal{S} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}_{>0}, \quad a \leq n+1, \quad b < \frac{2(n+1)}{M} \right\}.$$

If $\mathcal{S}$ contains numbers below M , choose $\varepsilon > 0$ smaller than $M/2$ and smaller than the minimum of the positive numbers $M - s$ for $s \in \mathcal{S}$, $s < M$. If it contains no such number, any $0 < \varepsilon < M/2$ will do. The ratio $> M - \varepsilon$ must then be at least M . Consequently, for every closed point $P \in U$, we obtain a rational curve E_P with

$$0 < -(E_P \cdot K_{X_P}) \leq n+1, \quad \frac{-(E_P \cdot K_{X_P})}{E_P \cdot H_P} \geq M.$$

This makes precise the argument of letting $\varepsilon \rightarrow 0$: we use finitely many numerical possibilities, without taking a limit of the curves themselves.

Furthermore, $H_P \cdot E_P \leq (n+1)/M$, uniformly in P . We can now reason as in Step 8 of Theorem 2.3.6, keeping both intersection numbers in the parameter space.

Choose $\ell > 0$ such that ℓH is very ample, and spread out the corresponding fixed embedding, after enlarging R and shrinking U if necessary, with

$$\mathcal{O}_{\mathcal{X}}(1) \simeq \mathcal{H}^{\otimes \ell}.$$

The preceding argument applies to every closed fibre of the resulting model. For each of the finitely many pairs of integers

$$1 \leq a \leq n+1, \quad b > 0, \quad \frac{a}{b} \geq M,$$

let $\mathcal{N}_{a,b}$ be the locus in $\text{Mor}_{\ell b}(\mathbb{P}_U^1, \mathcal{X})$ where

$$\deg g^* \omega_{\mathcal{X}/U}^{-1} = a, \quad \deg g^* \mathcal{H} = b.$$

The degrees of these pullback line bundles are locally constant in families of morphisms, so these loci are open and closed in their respective fixed-degree morphism spaces. Thus

$$\mathcal{N} = \coprod_{(a,b)} \mathcal{N}_{a,b}$$

is of finite type over U .

The normalization of each E_P gives a geometric point of $\mathcal{N}_P$. Chevalley's theorem therefore implies, as before, that $\mathcal{N}$ has a non-empty generic fibre. Base changing to the original algebraically closed field k , we obtain a non-constant morphism $g : \mathbb{P}_k^1 \rightarrow X$ with

$$a = \deg g^* \omega_X^{-1}, \quad b = \deg g^* \mathcal{O}_X(H), \quad 1 \leq a \leq n+1, \quad \frac{a}{b} \geq M.$$

Let $E = g(\mathbb{P}^1)_{\text{red}}$, which is rational by Lüroth's theorem, and let δ be the degree of g onto E . Then

$$-(K_X \cdot E) = \frac{a}{\delta}, \quad H \cdot E = \frac{b}{\delta}.$$

Hence

$$0 < -(K_X \cdot E) \leq n + 1, \quad \frac{-(K_X \cdot E)}{H \cdot E} = \frac{a}{b} \geq M,$$

which is the required conclusion. $\square$

From this theorem we see the strength of the bend-and-break technique: Frobenius makes the deformation space large in positive characteristic, and bounded degree allows us to recover a rational curve over the original field.

2.4 Mori's Cone Theorem

Recall that in Chapter 1 we introduced some concepts like cones, extremal rays and the nef cone. For a projective variety X , the cone of effective curves is a subcone of the $\mathbb{R}$ -vector space $N_1(X)_{\mathbb{R}}$. Related to the perfect pairing

$$N^1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R},$$

we have

$$\begin{aligned} \text{NE}_{\mathbb{Q}}(X) &= \left\{ \sum_i a_i [C_i] \mid C_i \subseteq X \text{ an irreducible curve, } a_i \in \mathbb{Q}_{\geq 0} \right\}, \\ \text{NE}(X) &= \left\{ \sum_i a_i [C_i] \mid C_i \subseteq X \text{ an irreducible curve, } a_i \in \mathbb{R}_{\geq 0} \right\}, \\ \overline{\text{NE}}(X) &= \text{the closure of } \text{NE}(X) \text{ in } N_1(X)_{\mathbb{R}}. \end{aligned}$$

Here every sum is finite. The nef cone lies in $N^1(X)_{\mathbb{R}}$ and is the dual cone of $\overline{\text{NE}}(X)$. Recall also that for a Cartier divisor D ,

$$D_{\geq 0} := \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) \geq 0\}, \quad D^{\perp} := \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) = 0\}.$$

We use the notation

$$\overline{\text{NE}}(X)_{D \geq 0} := \overline{\text{NE}}(X) \cap D_{\geq 0},$$

and similarly for > 0 , ≤ 0 and < 0 , also for $\mathbb{R}$ -Cartier divisors. Now we are ready to state the first result of Mori for smooth varieties of arbitrary dimension.

2.4A Cone Theorem in the Nonsingular Case

Theorem 2.4.1 (Cone Theorem). *Let X be a nonsingular projective variety such that K_X fails to be nef.*

- (1) *There are at most countably many rational curves $C_i \subseteq X$ such that $0 < -(C_i \cdot K_X) \leq \dim X + 1$, and*

$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{K_X \geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i].$$

- (2) *For any $\varepsilon > 0$ and any ample Cartier divisor H , we have*

$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{(K_X + \varepsilon H) \geq 0} + \sum_{i \in I(H, \varepsilon)} \mathbb{R}_{\geq 0}[C_i],$$

where $I(H, \varepsilon) = \{i \mid (C_i \cdot (K_X + \varepsilon H)) < 0\}$ is finite.

Proof. From Grothendieck's Theorem 2.1.11 we know that the scheme $\text{Mor}(\mathbb{P}_k^1, X)$ is the disjoint union of countably many schemes of finite type over k , hence has at most countably many irreducible components. Pick a component B , with its reduced structure, and consider the flat projective family

$$\pi : \mathbb{P}_k^1 \times B \longrightarrow B$$

and the universal evaluation map

$$f^{\text{univ}} : \mathbb{P}_k^1 \times \text{Mor}(\mathbb{P}_k^1, X) \longrightarrow X.$$

For any Cartier divisor D on X , the pullback of $\mathcal{O}_X(D)$ is a line bundle on $\mathbb{P}_k^1 \times B$ and is flat over B . Thus the function

$$b \longmapsto \chi(\mathbb{P}_b^1, f_b^* \mathcal{O}_X(D))$$

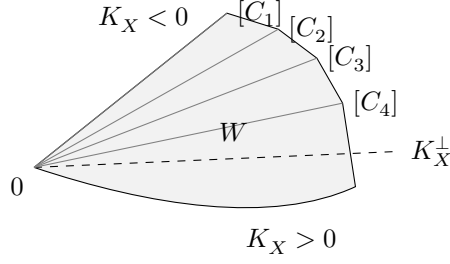
is locally constant. Since B is irreducible, it is connected, so this function is constant. By Riemann–Roch on $\mathbb{P}^1$, the intersection number $((f_b)_*[\mathbb{P}^1] \cdot D)$ is constant. Hence the *pushforward 1-cycles* parametrized by B are numerically equivalent. Every rational curve has a normalization map $\mathbb{P}^1 \rightarrow X$ whose pushforward is its curve cycle, so the numerical classes of rational curves also form an at most countable set.

In each numerical class γ represented by a rational curve with

$$0 < -(\gamma \cdot K_X) \leq \dim X + 1,$$

we pick one such curve C_i . This gives an at most countable index set I , and we define

$$W := \overline{\left(\overline{\text{NE}}(X)_{K_X \geq 0} + \sum_{i \in I} \mathbb{R}_{\geq 0}[C_i] \right)}.$$



Step 1: We aim to prove $W = \overline{\text{NE}}(X)$. Assume the contrary, so the natural inclusion $W \subseteq \overline{\text{NE}}(X)$ is strict. By separation of convex cones, followed by a small ample perturbation and rational approximation, we may choose a Cartier divisor D whose associated linear function $(D \cdot -)$ is positive on $W - \{0\}$ but negative somewhere in $\overline{\text{NE}}(X)$. Indeed, strict positivity can be checked on a compact slice cut out by an ample divisor, so it survives a sufficiently small perturbation; we then clear the denominators.

Let H be an ample Cartier divisor, and let $\mu > 0$ be the largest real number such that $H + \mu D$ is nef. This number is finite because D is negative somewhere in $\overline{\text{NE}}(X)$. The class $H + \mu D$ lies on the boundary of the nef cone. Thus, by Kleiman's criterion, there is a nonzero $z \in \overline{\text{NE}}(X)$ with

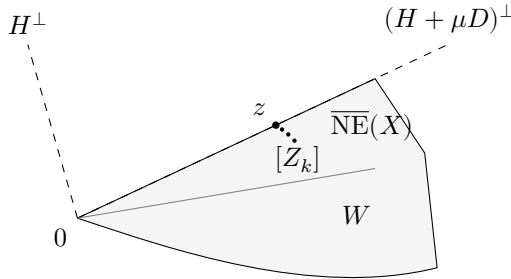
$$((H + \mu D) \cdot z) = 0.$$

Then $(D \cdot z) < 0$ because $(H \cdot z) > 0$, so $z \notin W$. Moreover $(K_X \cdot z) < 0$, since $\overline{\text{NE}}(X)_{K_X \geq 0} \subseteq W$.

Choose effective 1-cycles with rational coefficients $Z_k = \sum_j a_{kj} [Z_{kj}]$, where $a_{kj} > 0$, such that

$$Z_k \longrightarrow z \quad \text{as } k \longrightarrow \infty.$$

The situation is illustrated schematically below.



Fix a rational number $0 < \mu' < \mu$. The $\mathbb{Q}$ -Cartier divisor

$$L := H + \mu' D = \left(1 - \frac{\mu'}{\mu}\right) H + \frac{\mu'}{\mu} (H + \mu D)$$

is ample, being the sum of an ample and a nef $\mathbb{R}$ -divisor. For k sufficiently large, $-(Z_k \cdot K_X) > 0$. By Lemma 2.3.8,

$$\max_j \frac{-(Z_{kj} \cdot K_X)}{(Z_{kj} \cdot L)} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot L)} > 0.$$

Rearrange the indices so that the maximum is achieved for Z_{k0} . Apply Theorem 2.3.7 to Z_{k0} and L : formally, apply it to a positive Cartier multiple of L , then cancel that multiple in the ratio. We obtain a rational curve E_k such that

$$0 < -(E_k \cdot K_X) \leq \dim X + 1 \quad (2.2)$$

and

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot L)} \geq \frac{-(Z_{k0} \cdot K_X)}{(Z_{k0} \cdot L)} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot L)}.$$

By (2.2), E_k is numerically equivalent to one of the curves C_i . Thus $(E_k \cdot D) > 0$, by the choice of D , and consequently

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot H)} \geq \frac{-(E_k \cdot K_X)}{(E_k \cdot L)} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot L)}.$$

On the other hand, there is a positive constant $c \gg 0$ such that $K_X + cH$ is ample. Hence, for every such curve,

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot H)} < c.$$

But $-(z \cdot K_X) > 0$ and $((H + \mu D) \cdot z) = 0$. We can first choose a rational $\mu' < \mu$ sufficiently close to μ that

$$\frac{-(z \cdot K_X)}{(z \cdot (H + \mu' D))} > c.$$

Keep this μ' fixed. Since $Z_k \rightarrow z$, the same strict inequality holds with Z_k in place of z for k sufficiently large, a contradiction. This concludes Step 1.

Step 2: We prove part (2) of the theorem. Fix H and $\varepsilon > 0$. If $(C_i \cdot (K_X + \varepsilon H)) < 0$, then

$$(C_i \cdot H) < \frac{-(C_i \cdot K_X)}{\varepsilon} \leq \frac{\dim X + 1}{\varepsilon}.$$

By the second statement of Theorem 1.4.28, there are only finitely many such numerical classes. Since we chose one C_i per numerical class, there are only finitely many such C_i . Set

$$V_{H,\varepsilon} := \overline{\text{NE}}(X)_{(K_X + \varepsilon H) \geq 0} + \sum_{i \in I(H,\varepsilon)} \mathbb{R}_{\geq 0}[C_i].$$

This cone is closed. Indeed, in a convergent sequence $u_m + \sum_i a_{mi}[C_i]$ in this cone, with $a_{mi} \geq 0$, intersection with H bounds $H \cdot u_m$ and each a_{mi} . The set $\{u \in \overline{\text{NE}}(X) \mid H \cdot u \leq M\}$ is compact for every M . We can therefore pass to a subsequence on which all these terms converge, and the limit remains in $V_{H,\varepsilon}$.

Clearly $V_{H,\varepsilon} \subseteq \overline{\text{NE}}(X)$. It also contains $\overline{\text{NE}}(X)_{K_X \geq 0}$ and every ray $\mathbb{R}_{\geq 0}[C_i]$, and hence, being closed, it contains W . By Step 1, $W = \overline{\text{NE}}(X)$, which proves part (2).

Step 3: We prove part (1), including the removal of the closure in the definition of W . Recall the elementary finite-dimensional convexity fact that a closed convex cone with a compact base is generated by its extremal rays, using finite non-negative sums. To see its applicability here, intersect $\overline{\text{NE}}(X)$ with $H \cdot z = 1$. This is a compact convex set; the finite-dimensional Minkowski theorem and Carathéodory's theorem express every point as a finite convex combination of extreme points. These correspond to extremal rays of $\overline{\text{NE}}(X)$.

Let $R = \mathbb{R}_{\geq 0}r$ be a K_X -negative extremal ray. Choose $\varepsilon > 0$ small enough that $(K_X + \varepsilon H) \cdot r < 0$. By Step 2,

$$r = u + \sum_{i \in I(H,\varepsilon)} a_i[C_i], \quad u \in \overline{\text{NE}}(X)_{(K_X + \varepsilon H) \geq 0}, \quad a_i \geq 0.$$

Extremality implies that every nonzero summand belongs to R . Since $R - \{0\}$ is $(K_X + \varepsilon H)$ -negative, we must have $u = 0$. Thus some C_i generates R . Every remaining extremal ray is contained in $\overline{\text{NE}}(X)_{K_X \geq 0}$. The convexity fact above now gives

$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{K_X \geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i],$$

as required. In particular, each K_X -negative extremal ray is generated by a rational curve of anticanonical degree at most $\dim X + 1$. $\square$

Example 2.4.2 (Fano varieties). Let X be a Fano variety. Then $\overline{\text{NE}}(X) \subseteq N_1(X)_{\mathbb{R}}$ is a rational polyhedral cone, spanned by the classes of finitely many rational curves. Apply part (2) of the cone theorem with $H = -K_X$ and $\varepsilon = \frac{1}{2}$. Since $K_X + \varepsilon H = \frac{1}{2}K_X$ is negative on $\overline{\text{NE}}(X) - \{0\}$, the nonnegative part is $\{0\}$.

Now we introduce a key step: suitable extremal rays correspond to morphisms of a variety. This connects the cone of curves with more traditional geometric properties of the variety.

Definition 2.4.3. Let X be a normal projective variety and $F \subseteq \overline{\text{NE}}(X)$ an extremal face. A surjective morphism $\text{cont}_F : X \rightarrow Z$ to a normal projective variety is called the *contraction of F* if the following hold:

- (1) $\text{cont}_F(C)$ is a point for an irreducible curve $C \subseteq X$ if and only if $[C] \in F$, and F is the closed cone generated by these curve classes;

(2) $(\text{cont}_F)_* \mathcal{O}_X = \mathcal{O}_Z$; in particular, cont_F has connected fibres.

Remark 2.4.4. By Proposition 1.4.38, the contraction of F is unique up to isomorphism if it exists. Moreover, a morphism $g : X \rightarrow Z$ between normal projective varieties with $g_* \mathcal{O}_X = \mathcal{O}_Z$ is determined by the curves it contracts. Its relative cone $\text{NE}(g)$, as before, is generated by their classes; its closure is the corresponding extremal face of $\overline{\text{NE}}(X)$. Thus g is the contraction of this face. Now we give a lemma which will be used in Chapter 5.

Lemma 2.4.5. *Let X be a normal projective variety and let $r \subseteq \overline{\text{NE}}(X)$ be an extremal ray. Assume that X is $\mathbb{Q}$ -factorial and that the contraction of r between normal projective varieties exists,*

$$f = \text{cont}_r : X \longrightarrow Z.$$

Then one of the following holds:

- *f is birational and $\text{Ex}(f)$ is an irreducible divisor (we say that f is a **divisorial contraction**);*
- *f is birational and $\text{Ex}(f)$ has codimension at least two (we say that f is a **small contraction**);*
- *$\dim Z < \dim X$ (we say that f is **of fibre type**).*

Proof. If $\dim Z = \dim X$, the equality $f_* \mathcal{O}_X = \mathcal{O}_Z$ implies that f is birational. Thus the only assertion to prove is that $\text{Ex}(f)$ is irreducible when it contains a divisor.

Assume that $\text{Ex}(f)$ contains a prime divisor D . Since Z is normal, $f(D)$ has codimension at least two in Z . After replacing a projective embedding of Z by a sufficiently high Veronese embedding, choose a general integral hyperplane section H containing $f(D)$. Equivalently, choose a general member of a sufficiently positive linear system with base locus $f(D)$.

Let $H' \subseteq X$ be the strict transform of H . The proper morphism $H' \rightarrow H$ is surjective, so H' meets $f^{-1}(z)$ for every $z \in f(D)$. On the other hand, H' does not contain D . Thus for general $z \in f(D)$, the fibre $f^{-1}(z)$ meets H' but is not contained in it.

This fibre is connected. Since X is projective, we can choose an irreducible curve $C \subseteq f^{-1}(z)$ which meets H' but is not contained in it: a component of the fibre outside H' meets H' , and hyperplane sections through an intersection point give such a curve. By $\mathbb{Q}$ -factoriality, H' is $\mathbb{Q}$ -Cartier, so $(H' \cdot C) > 0$. Since C is contracted by f , we have $r = \mathbb{R}_{\geq 0}[C]$.

The Cartier divisor H has pullback

$$f^*H = H' + \sum_i a_i E_i,$$

where the E_i are the exceptional prime divisors that occur in the pullback, and $a_i > 0$. Locally these coefficients are the orders of vanishing of a local equation of H pulled back to X . Each E_i is $\mathbb{Q}$ -Cartier. Moreover, $(f^*H \cdot C) = (H \cdot f_*C) = 0$, hence

$$-(H' \cdot C) = \left(\sum_i a_i E_i \cdot C \right) < 0.$$

It follows that there is a divisor $E_i = E' \subseteq \text{Ex}(f)$ with $(E' \cdot C) < 0$.

For a proper birational morphism to a normal variety, every point of $\text{Ex}(f)$ lies on a positive-dimensional fibre, by Zariski's main theorem. The fibres are connected, so through any $x \in \text{Ex}(f)$ there is an irreducible contracted curve C' . Since f contracts the ray r ,

$$[C'] = \lambda[C] \quad \text{for some } \lambda \in \mathbb{R}_{>0}.$$

Therefore $(E' \cdot C') = \lambda(E' \cdot C) < 0$. An effective $\mathbb{Q}$ -Cartier divisor has nonnegative intersection with a curve not contained in its support, so $C' \subseteq E'$. Hence $x \in E'$. We conclude that $\text{Ex}(f) = E'$, as wanted. $\square$

In general, an extremal face need not admit a contraction to a projective variety, as the following example shows.

Example 2.4.6. For this example, work over $k = \mathbb{C}$. Let $g : X \rightarrow \mathbb{P}^2$ be the blow-up at twelve distinct very general points $p_1, \dots, p_{12}$ on a smooth plane cubic D . Let $C \subseteq X$ be the strict transform of D . By the canonical divisor formula for a blow-up [Har77, II, Exercise 8.5(b)],

$$\begin{aligned} (K_X \cdot C) &= \left(g^*K_{\mathbb{P}^2} + \sum_{i=1}^{12} E_i \right) \cdot C \\ &= \deg \mathcal{O}_D(-3) + \sum_{i=1}^{12} (E_i \cdot C) \\ &= -3 \times 3 + 12 = 3. \end{aligned}$$

Since $g^*D = C + \sum_{i=1}^{12} E_i$, we also have

$$C^2 = \left(g^*D - \sum_{i=1}^{12} E_i \right)^2 = 9 - 12 = -3.$$

Thus, by Example 1.4.31, $[C]$ spans an extremal ray r of $\overline{\text{NE}}(X)$. We claim that r has no contraction to a projective variety.

Indeed, suppose that such a contraction $\text{cont}_r : X \rightarrow Z$ exists. Since $r \neq \overline{\text{NE}}(X)$, the variety Z is not a point. Pulling back an ample line bundle on Z , we obtain a nontrivial line bundle on X

$$L = g^*\mathcal{O}_{\mathbb{P}^2}(d) \otimes \mathcal{O}_X \left(\sum_{i=1}^{12} a_i E_i \right)$$

whose restriction to C is trivial. Identifying C with D , this gives

$$\mathcal{O}_D(d) \otimes \mathcal{O}_D \left(\sum_{i=1}^{12} a_i p_i \right) \simeq \mathcal{O}_D.$$

Its degree is $3d + \sum_i a_i$, so a relation is possible only if $3d + \sum_i a_i = 0$. For every nonzero integer tuple satisfying this degree condition, at least one a_i is nonzero. Varying that p_i , multiplication by a_i on the elliptic curve is nonconstant. Thus the displayed relation holds on a proper closed subset of D^{12} . There are only countably many integer tuples. Choosing the points very generally avoids all these subsets, so the only possible relation has $d = a_1 = \dots = a_{12} = 0$. This contradicts the choice of L . Hence cont_r does not exist, as claimed.

2.4B Application: Minimal Models of Surfaces

For the rest of this section, all surfaces are smooth and projective over k . On a smooth surface, curves are Cartier divisors, and intersection identifies $N^1(X)_{\mathbb{R}}$ with $N_1(X)_{\mathbb{R}}$. We use this identification when writing z^2 for $z \in N_1(X)_{\mathbb{R}}$. To begin with, we need two lemmas.

Lemma 2.4.7. *If D is a Cartier divisor on a smooth projective surface X with $D^2 > 0$, then either $|nD| \neq \emptyset$ for all $n \gg 0$, or $|-nD| \neq \emptyset$ for all $n \gg 0$.*

Proof. By Riemann–Roch on a surface and Serre duality, dropping the nonnegative superabundance terms h^1 , we get

$$\begin{aligned} h^0(nD) + h^0(K_X - nD) &\geq \frac{n^2}{2} D^2 - \frac{n}{2} (D \cdot K_X) + \chi(\mathcal{O}_X), \\ h^0(-nD) + h^0(K_X + nD) &\geq \frac{n^2}{2} D^2 + \frac{n}{2} (D \cdot K_X) + \chi(\mathcal{O}_X). \end{aligned}$$

The right-hand sides grow quadratically. Set $B = h^0(2K_X)$. If both $H^0(K_X - nD)$ and $H^0(K_X + nD)$ are nonzero, multiplication by a fixed nonzero section of either line bundle injects the other space into $H^0(2K_X)$, since X is integral. Thus both dimensions are at most B . If one space is zero, the corresponding inequality already gives sections of nD or $-nD$.

For n sufficiently large that both right-hand sides exceed B , it follows that $h^0(nD) > 0$ or $h^0(-nD) > 0$. These signs cannot vary with n : for an ample divisor H , a nonzero effective divisor has positive intersection with H , whereas $(nD \cdot H)$ and $(-nD \cdot H)$ have opposite signs. This also shows $(D \cdot H) \neq 0$. Hence one fixed sign works for all $n \gg 0$. $\square$

Corollary 2.4.8. *Let X be a smooth projective surface with an ample Cartier divisor H . The set*

$$Q := \{z \in N_1(X)_{\mathbb{R}} \mid z^2 > 0\}$$

has two connected components,

$$Q^+ := \{z \in Q \mid (z \cdot H) > 0\}, \quad Q^- := \{z \in Q \mid (z \cdot H) < 0\}.$$

Furthermore, $Q^+ \subseteq \overline{NE}(X)$.

Proof. We have the intersection form

$$(-, -) : N_1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}, \quad (z_1, z_2) \longmapsto (z_1 \cdot z_2).$$

Write $\rho = \rho(X)$. By the Hodge index theorem,

$$N_1(X)_{\mathbb{R}} = \mathbb{R}[H] \oplus H^{\perp}, \quad H^{\perp} = \{z \in N_1(X)_{\mathbb{R}} \mid (z \cdot H) = 0\},$$

where the intersection form is positive definite on the one-dimensional space $\mathbb{R}[H]$ and negative definite on the $(\rho - 1)$ -dimensional space $H^{\perp}$.

More precisely, for a basis $\{D_i\}_{i=1}^{\rho}$, let $M = (D_i \cdot D_j)_{i,j}$. Writing $z_1 = \sum_i x_i D_i$ and $z_2 = \sum_i y_i D_i$, we have

$$(z_1 \cdot z_2) = x^T M y.$$

The real symmetric matrix M can be orthogonally diagonalized. After also rescaling the basis vectors, Sylvester's law of inertia gives a basis $\{e_i\}_{i=1}^{\rho}$, orthogonal for the intersection form, in which the matrix is

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 \end{pmatrix}.$$

We may take $e_1 = [H]/\sqrt{H^2}$ and choose the remaining basis vectors in $H^{\perp}$, so

$$[H] = (\sqrt{H^2}, 0, \dots, 0).$$

For $z = \sum_i x_i e_i \in Q$, we have

$$x_1^2 - \sum_{i \geq 2} x_i^2 > 0, \quad \text{or equivalently} \quad |x_1| > \sqrt{\sum_{i \geq 2} x_i^2}.$$

This gives the two open convex, hence connected, components

$$Q^+ = \left\{ x_1 > \sqrt{\sum_{i \geq 2} x_i^2} \right\}, \quad Q^- = \left\{ x_1 < -\sqrt{\sum_{i \geq 2} x_i^2} \right\}.$$

Since $(z \cdot H) = x_1 \sqrt{H^2}$, these are exactly the components in the statement.

For the second assertion, first take a rational class $[D] \in Q^+$. After clearing denominators, we may assume that D is Cartier. By Lemma 2.4.7, either nD or

$-nD$ is linearly equivalent to an effective divisor for $n \gg 0$. An effective nonzero divisor has positive intersection with H , so $(D \cdot H) > 0$ excludes the negative sign. Hence $[D] = \frac{1}{n}[nD] \in \overline{\text{NE}}(X)$. Rational classes are dense in the open set Q^+ , and $\overline{\text{NE}}(X)$ is closed. Thus $Q^+ \subseteq \overline{\text{NE}}(X)$. In particular, every point of Q^+ is an interior point of $\overline{\text{NE}}(X)$. $\square$

Now we give the main result in this section (even in this chapter!), classifying the possible contractions on surfaces when K_X fails to be nef.

Theorem 2.4.9 (MMP for Surfaces). *Let X be a smooth projective surface and $R \subseteq \overline{\text{NE}}(X)$ a K_X -negative extremal ray. Then the contraction morphism $\text{cont}_R : X \rightarrow Z$ exists and is of one of the following types:*

- (1) Z is a smooth projective surface and X is obtained from Z by blowing up a closed point; $\rho(Z) = \rho(X) - 1$.
- (2) Z is a smooth projective curve and X is a geometrically ruled surface over Z , that is, a $\mathbb{P}^1$ -bundle; $\rho(X) = 2$.
- (3) Z is a point, $\rho(X) = 1$ and $-K_X$ is ample. (In fact, $X \simeq \mathbb{P}^2$, but this requires an additional argument.)

Proof. Step 1: By the cone theorem, Theorem 2.4.1 and its proof, we may choose an irreducible rational curve $C \subseteq X$ with $R = \mathbb{R}_{\geq 0}[C]$ and

$$0 < -(K_X \cdot C) \leq \dim X + 1 = 3.$$

Hence $(K_X \cdot C) \in \{-1, -2, -3\}$. Recall that the arithmetic genus is defined by

$$p_a(C) := 1 - \chi(C, \mathcal{O}_C) = h^1(C, \mathcal{O}_C),$$

since C is integral and proper. As a consequence of Riemann–Roch for surfaces, we have the adjunction formula [Har77, V, Exercise 1.3(a)]:

$$2p_a(C) - 2 = (C \cdot (C + K_X)) = C^2 + (K_X \cdot C).$$

Since $p_a(C) \geq 0$ (notice that C may be singular!), this implies $C^2 \geq -2 - (K_X \cdot C) \geq -1$. Equality $C^2 = -1$ holds if and only if $(K_X \cdot C) = -1$ and $p_a(C) = 0$. In that case $C \simeq \mathbb{P}^1$, so C is a (-1) -curve. Castelnuovo’s contraction theorem contracts exactly C to a smooth point on a smooth projective surface Z . This is the contraction of R , and $\rho(Z) = \rho(X) - 1$, giving case (1).

Step 2: It remains to deal with $C^2 > 0$ and $C^2 = 0$. Suppose first that $C^2 > 0$. By Corollary 2.4.8, $[C] \in Q^+$ is an interior point of $\overline{\text{NE}}(X)$. Since $[C]$ also spans an extremal ray, this forces $\rho(X) = 1$ and $\overline{\text{NE}}(X) = R$. Thus $X \rightarrow \text{Spec } k$ is the contraction of R . By assumption $(K_X \cdot C) < 0$, so $-K_X$ is positive on $\overline{\text{NE}}(X) - \{0\}$

and is ample by Kleiman's criterion, Theorem 1.4.28. For the further conclusion $X \simeq \mathbb{P}^2$, see the classification of smooth del Pezzo surfaces, or [Mat02, I-4-4].

Step 3: Now consider $C^2 = 0$. Adjunction gives $2p_a(C) - 2 = (K_X \cdot C) < 0$, so $\overline{p_a(C)} = 0$, $(K_X \cdot C) = -2$ and $C \simeq \mathbb{P}^1$.

We prove that a multiple of C gives the desired contraction. For a fixed ample divisor H , $(K_X - mC) \cdot H < 0$ for $m \gg 0$, since $C \cdot H > 0$. Thus, by Serre duality,

$$H^2(X, \mathcal{O}_X(mC)) \simeq H^0(X, \mathcal{O}_X(K_X - mC))^\vee = 0 \quad (m \gg 0).$$

Riemann–Roch on the surface now gives

$$h^0(X, \mathcal{O}_X(mC)) \geq \chi(X, \mathcal{O}_X(mC)) = -\frac{m}{2}(K_X \cdot C) + \chi(\mathcal{O}_X) = m + \chi(\mathcal{O}_X),$$

which grows linearly. In particular, for m sufficiently large, $h^0(X, \mathcal{O}_X(mC)) \geq 2$.

Write the complete linear system as $|mC| = F + |M|$, where F is its fixed part. Since mC itself is a member of $|mC|$, we have $F \leq mC$ as divisors. Because C is irreducible, this implies $F = kC$ for an integer $0 \leq k < m$. Removing the fixed part gives the complete linear system

$$|M| = |(m - k)C|.$$

Set $m' = m - k > 0$. Then $|m'C|$ has no fixed component and has dimension at least one.

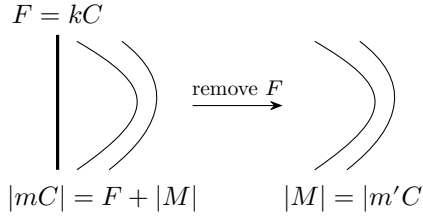


Figure 2.10: Removing the fixed component from $|mC|$ (schematic).

Next we verify that $|m'C|$ is base point free. Since it has no fixed component, we may choose $D \in |m'C|$ which does not contain C . Then

$$(D \cdot C) = m'C^2 = 0,$$

so D is disjoint from C . But $m'C$ is another member of the same system. These two members have disjoint supports, and hence $|m'C|$ has no base points.

Let $\phi : X \rightarrow Y \subseteq \mathbb{P}_k^N$ be the corresponding morphism. Its image has dimension one: it is nonconstant, and a two-dimensional image would imply $(m'C)^2 > 0$. Take the Stein factorization

$$\phi : X \xrightarrow{f} Z \xrightarrow{h} Y \subseteq \mathbb{P}_k^N, \quad Z = \text{Spec}_Y(\phi_* \mathcal{O}_X).$$

Here $f_*\mathcal{O}_X = \mathcal{O}_Z$, h is finite, and Z is a normal projective curve, hence smooth. We have not yet assumed that $f = \text{cont}_R$.

Put $A = h^*\mathcal{O}_Y(1)$, so $f^*A \simeq \mathcal{O}_X(m'C)$. By the projection formula and $f_*\mathcal{O}_X = \mathcal{O}_Z$, the section with zero divisor $m'C$ is the pullback of a section of A . Its zero divisor on Z is supported at one point z_0 , since its pullback is supported on the irreducible curve C . Hence the scheme-theoretic fibre over z_0 is aC for some integer $a > 0$.

The morphism f is flat, since it is a dominant morphism from a smooth surface to a smooth curve. All its fibres are numerically equivalent as divisors. A general fibre G is smooth and connected by generic smoothness in characteristic zero and the Stein factorization. Thus $G \equiv aC$, and adjunction gives

$$2g(G) - 2 = G^2 + (K_X \cdot G) = a(K_X \cdot C) = -2a.$$

Since $g(G) \geq 0$ and $a > 0$, we get $a = 1$ and $g(G) = 0$. Therefore every fibre has numerical class $[C]$.

Let $\sum_i a_i C_i$ be any fibre, with $a_i \in \mathbb{Z}_{>0}$ and C_i its distinct irreducible components. We now have

$$\sum_i a_i [C_i] = [C] \in R.$$

Since R is extremal, every $[C_i]$ lies in R . Consequently $C_i^2 = 0$ and $(K_X \cdot C_i) < 0$. Adjunction gives

$$2p_a(C_i) - 2 = (K_X \cdot C_i) < 0,$$

so $p_a(C_i) = 0$, $C_i \simeq \mathbb{P}^1$ and $(K_X \cdot C_i) = -2$. It follows that

$$-2 = (K_X \cdot C) = \left(K_X \cdot \sum_i a_i C_i \right) = -2 \sum_i a_i.$$

Thus $\sum_i a_i = 1$, and every fibre is irreducible, reduced, and isomorphic to $\mathbb{P}^1$.

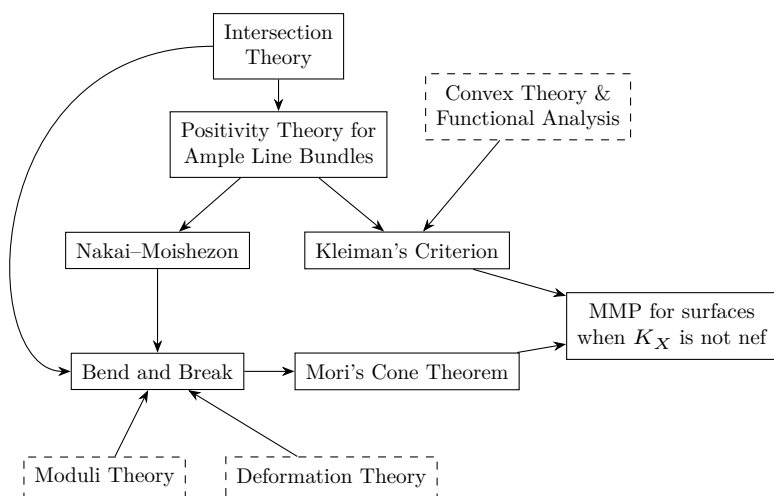
Moreover, an irreducible curve E is contracted by f if and only if $[E] \in R$. One direction follows from the description of the fibres. For the other, if $[E] \in R$, then $(f^*A \cdot E) = m'(C \cdot E) = 0$; since A is ample on Z , E must be contracted. Hence $f = \text{cont}_R$.

Finally, Tsen's theorem gives a $k(Z)$ -rational point on the genus-zero generic fibre. This yields a section over an open subset of Z , which extends over all of the smooth curve Z by properness. Thus X is a geometrically ruled surface over Z . By [Har77, V, Proposition 2.2], it is the projectivization of a rank-two locally free sheaf on Z :

$$X \simeq \mathbb{P}_Z(\mathcal{E}).$$

In particular, $\rho(X) = \rho(Z) + 1 = 2$, as also follows from [Har77, II, Exercise 7.9(a)]. This is case (2). $\square$

Summary:



Dashed boxes: only basic tools needed.
Solid boxes: main results to understand.

Figure 2.11: Knowledge framework.

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Additional references for the revised passages. For graphs and the construction of the morphism scheme, see B. Conrad’s course notes, written by T. Feng and A. Landesman, *Math 249B: Alterations*, Remark 8.12 and the proof of Lemma 9.6 (pp. 49 and 56). For the bounded-degree lifting argument and the convexity used in the cone theorem, see O. Debarre, *Bend and Break*, pp. 24–25 and Section 9.1. For ampleness in families, constructible images, and the final ruled-surface argument, see the Stacks Project, Tag 0D3A, Tag 054H, and Tag 03RD. References to numbered results in other chapters are to the original reading notes.